

An Explainable Deep Learning Framework for Predicting Traffic Accident Risk and Identifying Critical Factors Using CNN–BiLSTM–Attention and SHAP

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CERTIFICATE

This is to certify that the project that is entitled with the name “An Explainable Deep Learning Framework for Predicting Traffic Accident Risk and Identifying Critical Factors Using CNN–BiLSTM–Attention and SHAP” is a Bonafide work done by **Kallam Thirupatamma (22471A0526), Biraka Velangini Rani (22471A0504), Yenumala Mythri (22471A0569)** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in the Department of **COMPUTER SCIENCE AND ENGINEERING** during 2025-2026.

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PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

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PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

Project Course Outcomes (CO'S):

CO421.1: Analyse the System of Examinations and identify the problem.

CO421.2: Identify and classify the requirements.

CO421.3: Review the Related Literature

CO421.4: Design and Modularize the project

CO421.5: Construct, Integrate, Test and Implement the Project.

CO421.6: Prepare the project Documentation and present the Report using appropriate method.

Course Outcomes – Program Outcomes mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
C421.1		✓										✓		
C421.2	✓		✓		✓							✓		
C421.3				✓		✓	✓	✓				✓		
C421.4			✓			✓	✓	✓				✓	✓	
C421.5					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
C421.6									✓	✓	✓	✓	✓	

Course Outcomes – Program Outcome correlation

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
C421.1	2	3										2		
C421.2			2		3							2		
C421.3				2		2	3	3				2		
C421.4			2			1	1	2				3	2	
C421.5					3	3	3	2	3	2	2	3	2	1
C421.6									3	2	1	2	3	

Note: The values in the above table represent the level of correlation between CO's and PO's:

1. Low level
2. Medium level
3. High level

Project mapping with various courses of Curriculum with Attained PO's:

Name of the course from which principles are applied in this project	Description of the device	Attained PO
C2204.2, C22L3.2	Gathering the requirements and defining the problem, plan to develop a model for predicting traffic accident risk using CNN–BiLSTM–Attention.	PO1, PO3, PO8
CC421.1, C2204.3, C22L3.2	Each and every requirement is critically analyzed, and key traffic, weather, temporal, and spatial factors are identified.	PO2, PO3, PO8
CC421.2, C2204.2, C22L3.3	Logical design is done using deep learning principles and attention mechanisms, involving team collaboration for CNN–BiLSTM–Attention architecture.	PO3, PO5, PO9, PO8
CC421.3, C2204.3, C22L3.2	Each and every module is tested, integrated, and evaluated using metrics like Accuracy, Precision, Recall, F1-score, and Jaccard Index.	PO1, PO5, PO8
CC421.4, C2204.4, C22L3.2	Documentation is done by all team members, including data preprocessing, modeling, training, evaluation, and SHAP explainability analysis.	PO10, PO8
CC421.5, C2204.2, C22L3.3	Each phase of the work in the group is presented periodically, covering workflow, results, and insights	PO8, PO10, PO11
C2202.2, C2203.3, C1206.3, C3204.3, C4110.2	Implementation is done, and the system predicts accident risk based on input features; future updates can include additional datasets or factors.	PO4, PO7, PO8
C32SC4.3	The physical design includes a dashboard/web application to visualize predicted accident risk and key contributing factors	PO5, PO6, PO8

ABSTRACT

Accurate estimation of accident risk and the identification of critical contributing factors are critical for improving traffic safety and road fatality reduction. This paper introduces a new deep learning architecture that combines Convolutional Neural Networks (CNN), Bidirectional Long Short-Term Memory (BiLSTM), and Attention to capture both spatial and temporal trends in traffic accident data. To provide interpretability and credibility for decision-making, the model is augmented further with SHAP (Shapley Additive explanations) to identify the most significant features responsible for accident severity. The model is tested on two real datasets: the US Accident Dataset (March 2023) and the UK STATS19 dataset, following rigorous preprocessing, feature normalization, and class balancing using SMOTE Tomek and RandomOverSampler methods. The suggested CNN–BiLSTM–Attention model presents high predictive accuracy, MAE of 0.137 (US) and 0.179 (UK), far exceeding conventional models such as Logistic Regression. Through SHAP analysis, the framework reveals transparent insights regarding the key factors determining accident severity, and the framework is both precise and interpretable. This paper presents an efficient and scalable solution for real time traffic risk prediction and traffic authority decision support.

INDEX

S.NO	CONTENT	PAGE NO
1	INTRODUCTION	1
	1.1 MOTIVATION	3
	1.2 PROBLEM STATEMENT	5
	1.3 OBJECTIVE	6
2	LITERATURE SURVEY	7
3	SYSTEM ANALYSIS	
	3.1 EXISTING SYSTEM	13
	3.1.1 DISADVANTAGES OF THE EXISTING SYSTEM	16
	3.2 PROPOSED SYSTEM	17
	3.3 FEASIBILITY STUDY	19
	3.4 USING COCOMO MODEL	20
4	SYSTEM REQUIREMENTS	
	4.1 SOFTWARE REQUIREMENT	22
	4.2 REQUIREMENT ANALYSIS	22
	4.3 HARDWARE REQUIREMENTS	24
	4.4 SOFTWARE	24
	4.5 SOFTWARE DESCRIPTION	25
5	SYSTEM DESIGN	
	5.1 SYSTEM ARCHITECTURE	27
	5.1.1 DATASET	28
	5.1.2 DATA PREPROCESSING	31
	5.1.3 FEATURE EXTRACTION	32
	5.1.4 MODEL BUILDING	33
	5.1.5 CLASSIFICATION	40
	5.2 MOUDELS	42
	5.3 UML DIAGRAMS	46
6	IMPLEMENTATION	
	6.1 MODEL IMPLEMENTATION	50
	6.2 CODING	54
7	TESTING	

7.1		
	UNIT TESTING	68
7.2		
	INTEGRATION TESTING	69
7.3		
	SYSTEM TESTING	71
8	RESULT ANALYSIS	76
9	OUTPUT SCREENS	81
10	CONCLUSION	84
11	FUTURE SCOPE	85
12	REFERENCES	86

LIST OF FIGURES

S.NO	FIGURE DESCRIPTION	PAGE NO
1.	FIG3.1 FLOW CHART OF EXISTING SYSTEM FOR TRAFFIC ACCIDENT RISK PREDICTION	14
2.	FIG 3.2 FLOWCHART OF PROPOSED SYSTEM	17
3.	FIG 5.1 IMAGE AFTER APPLYING THE PREPROCESSING TECHNIQUE	32
4.	FIG 5.2 FEATURE IMPORTANCE BEFORE vs AFTER	33
5.	FIG 5.3 CNN MODEL ARCHITECTURE	35
6.	FIG 5.4 BIDIRECTIONAL LONG SHORT-TERM MEMORY ARCHITECTURE	36
7.	FIG 5. 5 CNN–BiLSTM–ATTENTION HYBRID ARCHITECTURE	38
8.	FIG 5.6 DESIGN OVERVIEW	47
9.	FIG 5.8 UMLClass Diagramofthe Proposed CNN–BiLSTM–Attention	48
10.	FIG 7.1 STATUS FATAL ACCIDENT	72
11.	FIG 7.2 STATUS SERIOUS ACCIDENT	73
12.	FIG 7.3 STATUS SLIGHT ACCIDENT	74
13.	FIG 7.4 BULK FILE PREDICTS	75
14.	FIG 8.1 ACCURACY COMPARISON ON DIFFERENT MODELS	76
15.	FIG 8.2 CONFUSION MATRIX	78
16.	FIG 8.3 TOP 15 FEATURES BY SHAP	79
17.	FIG 9.1 HOME PAGE	81
18.	FIG 9.2 ABOUT PAGE	82
19.	FIG 9.3 PREDICTION PAGE	82
20.	FIG 9.4 RESULT PAGE	83
21.	FIG 9.5 HISTORY PAGE	83

List of Tables

S.NO	CONTENT	PAGE NO
1	TABLE 1. DATASET DESCRIPTION	29

1.INTRODUCTION

Road traffic crashes are among the major causes of human misery, injuries, and deaths globally, and they impose a major public health and economic cost. The World Health Organization (WHO) indicates that an estimated 1.35 million individuals die annually because of road crashes, while millions of others experience non-fatal injuries often leading to permanent disabilities [1]. The effects of road accidents go beyond human fatalities to include extensive social and economic consequences, such as a loss in productivity, health costs, and damage to infrastructure. The severity of the accident depends on a multifaceted combination of factors like the behavior of the driver, the type of vehicle, road design, atmospheric conditions, and chronological factors like time or congestion [2].

Road traffic accidents are a serious concern for health and safety in India, with the nation contributing to almost 11% of global road accident fatalities [3]. Government statistics say that over 150,000 people lose their lives and over 450,000 are injured every year on Indian roads, with higher impacts on the younger age groups of 18–45 years. The increasing trend in road accidents in urban and rural locations can be linked to the causes including speeding, bad road conditions, distraction while driving, and weak enforcement of traffic laws. The financial burden caused by road traffic accidents in India alone is estimated to be close to 3% of its Gross Domestic Product (GDP) [4]. Such disturbing figures emphasize the critical importance of effective and scalable means of anticipating and preventing accident risks.

Current accident risk assessment techniques mostly utilize statistical models or traditional machine learning algorithms. Although effective, these approaches are not equipped to represent the non-linear, high-dimensional, and dynamic traffic data, particularly with spatial and temporal dependencies playing a key role [5]. In addition, most of these models are "black boxes" that provide little or no explanation of what is causing accident severity. Such a lack of transparency is hindering their practical use in high-stakes fields such as transportation safety, where decision-makers need accurate predictions as well as transparent explanations in support of policy-making and resource distribution.

With the development of artificial intelligence (AI) and deep learning, there is great potential to address these challenges. Deep learning models, especially Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), have shown incredible success in numerous applications by efficiently extracting intricate spatial and temporal features from big data [6]. In order to solve the shortcomings of current methods, this paper proposes a deep learning architecture hybrid that combines CNNs for extracting spatial features, BiLSTM networks for temporal sequence modeling, and an attention mechanism that allows the model to pay attention to the most salient input features while making predictions.

Apart from predictive accuracy, this research highlights explainability through the inclusion of SHAP (Shapley Additive explanations), an explainable AI method. SHAP offers both global and local interpretability through the highlighting of significant features contributing to the severity of accidents, thus providing actionable information for traffic authorities, policymakers, and emergency responders [7]. The suggested CNN–BiLSTM–Attention–SHAP approach not only improves the accuracy of accident severity prediction but also guarantees transparency, reliability, and feasibility for real-world implementation.

By facilitating the identification of significant contributing factors like weather, road type, time of day, and driver-related variables, the model enhances data-driven decision-making in accident prevention and traffic management. In the end, this research intends to enhance emergency response strategies, optimize resource allocation, and assist in the design of safer, smarter transport systems.

1.1 Motivation

Road traffic accidents remain a worldwide health epidemic, causing enormous human, social, and economic losses. In spite of improvements in vehicle safety technologies and roadway infrastructure, the incidence and severity of traffic crashes are still very high, especially in low- and middle-income nations like India. Thousands lose their lives every year, and millions get injured and incur long-term disabilities, imposing a huge cost on healthcare systems as well as national economies [1].

One of the most critical challenges is the timely and precise forecasting of accident severity, which can be a key factor in casualty minimization by means of efficient emergency response and preventive traffic management measures. Most conventional approaches to accident analysis depend on subjective evaluation of crash reports or elementary statistical models, which tend to ignore intricate interdependencies between various factors like driver actions, road conditions, traffic flow, weather, and time of day [2]. Not only are such approaches labor-intensive, but also subject to subjectivity and inhomogeneities, thereby restricting their applicability in practice.

With the general usage of smart transportation systems and access to large volumes of traffic accident data, there is increasing potential to apply artificial intelligence (AI) and deep learning methods for accident severity prediction. State-of-the-art models like Convolutional Neural Networks (CNNs) and Bidirectional Long Short-Term Memory (BiLSTM) networks can learn latent spatial-temporal features from traffic data, making the prediction more accurate and scalable than traditional methods [3].

But predictive accuracy at high levels is not enough in high-stakes applications such as road traffic safety. Interpretability is also crucial since policymakers and stakeholders need to grasp the underlying factors of accident severity in order to create focused interventions. It is here that explainable AI (XAI) methods such as

SHAP (Shapley Additive explanations) play a crucial role, providing transparent insight into model predictions and building trust in AI-driven decision-making [4].

The driving force behind this research is thus to address both the requirements of precise accident severity prediction and model interpretability. Through the construction of a hybrid approach combining CNNs, BiLSTMs, attention, and SHAP, this research seeks to produce a robust, understandable, and scalable traffic accident analysis solution. Such a model not only enhances diagnostic performance and speeds up response times but also equips traffic authorities with informative actionable, leading to safer roads, decreased deaths, and improved transportation infrastructure.

1.2 Problem Statement

Road traffic accidents are complex phenomena influenced by multiple interacting factors such as driver behavior, road infrastructure, weather conditions, vehicle type, and temporal attributes like time of day and traffic density. Traditional analytical and machine learning models are often insufficient to capture these intricate spatial and temporal relationships, resulting in limited predictive accuracy and interpretability.

Most existing accident prediction systems rely on static data and linear assumptions, making them incapable of modeling the dynamic and nonlinear dependencies that govern real-world traffic events. Additionally, many deep learning frameworks, though powerful in prediction, act as “black boxes,” offering limited transparency regarding the rationale behind their outputs. This lack of interpretability hinders their practical adoption in safety-critical applications such as transport planning and accident prevention.

Therefore, the key problem addressed in this research is the lack of an accurate, interpretable, and scalable framework for predicting traffic accident severity using real-world heterogeneous datasets. The system must not only predict accident severity with high precision but also provide a clear understanding of the significant factors contributing to such events.

The proposed Explainable Deep Learning Framework (CNN–BiLSTM–Attention–SHAP) seeks to overcome these limitations by integrating:

CNN for spatial feature extraction, BiLSTM for temporal sequence learning,

Attention for identifying influential features, and SHAP for post-hoc explainability.

This hybrid model aims to deliver an intelligent, interpretable, and data-driven solution that bridges the gap between accuracy and transparency. By combining predictive modelling and explainable AI, the framework supports evidence-based decision-making for traffic management authorities, helping them to identify high-risk factors, plan preventive measures, and enhance road safety. The outcome of this research is expected to contribute significantly to the development of safer, smarter, and more sustainable transportation systems.

1.3 Objective

The main objective of this project is to develop an Explainable Deep Learning Framework for predicting road traffic accident severity with high accuracy and transparency. The proposed model integrates Convolutional Neural Networks (CNN), Bidirectional Long Short-Term Memory (BiLSTM), an Attention Mechanism, and SHAP (Shapley Additive Explanations) to effectively capture spatial-temporal dependencies and provide interpretable results.

The specific objectives are outlined as follows:

1. To preprocess and prepare real-world accident datasets
 - a. Collect data from the UK STATS19 and US Accident datasets.
 - b. Perform data cleaning, encoding, normalization, and handle class imbalance using techniques such as SMOTE-Tomek or RandomOverSampler.
2. To design a hybrid deep learning model
 - a. Use CNN for extracting spatial patterns and BiLSTM for capturing sequential and temporal dependencies in traffic data.
 - b. Integrate an Attention Mechanism to emphasize the most significant factors influencing accident severity.
3. To ensure model interpretability through explainable AI
 - a. Incorporate SHAP to interpret both global and local feature contributions.
 - b. Identify and visualize the top 15 key factors contributing to accident severity.
4. To evaluate model performance and compare with existing methods
 - a. Assess the model using metrics such as Accuracy, Precision, Recall, F1 - Score, and Mean Absolute Error (MAE).
 - b. Compare performance with baseline models such as Logistic Regression.
5. To support decision-making in transportation safety
 - a. Provide interpretable visual insights for policymakers and traffic authorities.
 - b. Enable data-driven strategies to reduce accident risks and enhance road safety.

2. LITERATURE SURVEY

Traffic accident risk prediction has gained increasing attention with the growth of intelligent transportation systems and the demand for accurate and interpretable safety solutions. Traditional Machine Learning models such as Logistic Regression, SVM, and Random Forest have been used to estimate accident severity, but they struggle to capture complex spatial-temporal dependencies in real-world traffic data. Deep Learning techniques, particularly Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), effectively learn spatial and temporal features from heterogeneous datasets, improving prediction accuracy. Hybrid models combining CNN and BiLSTM architectures further enhance performance by capturing both spatial patterns and temporal trends. Attention mechanisms enable models to focus on critical factors like weather and driver behaviour, while Explainable AI methods, especially SHAP, improve interpretability by identifying the most influential risk features. These developments have led to the emergence of advanced frameworks such as CNN-BiLSTM-Attention-SHAP, offering both high predictive accuracy and transparency for practical traffic safety applications.

Y. Wang et al. [1] developed a hybrid deep learning model integrating GRU and AdaBoost to predict tunnel traffic accidents using multi-heterogeneous data such as visibility, weather, and traffic flow. The model achieved an accuracy of 84.59%, outperforming conventional GRU networks by effectively fusing multi-source data and adjusting feature weights adaptively. However, the framework lacked explainability mechanisms like SHAP or LIME, limiting its interpretability and applicability for real-world safety decision-making.

Tian et al. [2] proposed a multimodal deep learning framework that integrates aerial imagery with contextual features such as road topology, traffic flow, and surrounding building layouts using segmentation models like DeepLabV3+, UNet++, and SegFormer. The inclusion of a spatial visibility map enhanced intersection-level accident risk identification by addressing occlusion effects. However, the model lacked temporal dynamics and interpretability components, which are crucial for understanding evolving risk patterns and supporting data-driven traffic safety policies.

R. Zepf et al. [3] analyzed virtual traffic simulations incorporating human error and Safety Performance Indicators (SPIs) to evaluate road risk in controlled environments. The study provided valuable insights into driver behavior and accident causation through simulated scenarios. However, it lacked real-world data validation and deep learning integration, limiting its applicability for real-time traffic accident prediction and decision-making systems.

Similarly, V. Chandrasekhar et al. [4] developed a clustering-based anomaly detection system for real-time traffic data analysis, leveraging live sensor inputs to enhance urban congestion management and dynamic traffic control. While effective in detecting irregular traffic patterns, the model was restricted to anomaly identification and did not assess accident severity levels or incorporate interpretability mechanisms necessary for understanding underlying risk factors.

Shim et al. [5] developed a dual-layer risk assessment framework for autonomous vehicles that integrates static and dynamic risk maps to support real-time collision avoidance maneuvers. The model effectively quantified risk for AV navigation scenarios but was restricted by its limited applicability to general traffic conditions and the absence of explainability mechanisms, reducing its interpretive value for broader transportation safety analysis.

Thakur et al. [6] introduced Safe Trax, an IoT-enabled real-time collision alert system that utilizes GPS, vibration, and ultrasonic sensors to detect potential crashes and notify users through mobile alerts. While the system effectively provided proximity-based warnings, it lacked predictive deep learning components and analytical modeling capabilities, thereby limiting its use for comprehensive accident risk prediction and severity analysis.

Jeong et al. [7] proposed a visual attention divergence model that predicts potential accidents by analyzing discrepancies between a driver's gaze behavior and the vehicle's motion vectors. The approach effectively identifies distraction-induced risks using vision-based analysis from in-vehicle cameras. However, it is limited to image-level features and lacks integration with structured traffic datasets or explainability mechanisms, reducing its interpretive power for large-scale accident prediction.

Liang et al. [8] developed a self-supervised multiscale convolutional neural network (CNN) framework to predict crash risk using satellite imagery and contrastive learning. The model effectively captured spatial and contextual patterns from large-scale aerial data without requiring manual labeling. However, it was restricted to static imagery and lacked integration with temporal traffic dynamics or explainable AI components, limiting its applicability for real-time and interpretable accident risk prediction.

Gao et al. [9] proposed a probabilistic risk assessment framework employing Extended Kalman Filters and Gaussian modeling to evaluate uncertainties in driver behavior, road geometry, and sensor measurements. The system estimated collision probabilities under diverse driving conditions and dynamic maneuvers such as lane changes and emergency braking. While effective for short-term trajectory-based risk assessment, it lacked scalability and explainability features, limiting its use for broader infrastructure-level accident prediction.

Zhao et al. [10] implemented a hybrid CNN–BiLSTM framework for traffic accident severity prediction, where CNN extracted spatial correlations from road and environmental features, and BiLSTM modeled temporal dependencies across time-series data. The hybrid model outperformed traditional machine learning methods, demonstrating superior accuracy in capturing complex spatial–temporal relationships. However, it lacked interpretability mechanisms, making it difficult to identify the contribution of individual features to severity outcomes.

Yang et al. [11] enhanced the CNN–BiLSTM architecture by integrating an Attention Mechanism to identify and prioritize key features such as weather, lighting, and traffic density in accident severity prediction. The attention layer improved model accuracy and interpretability by emphasizing the most influential inputs during prediction. However, the framework provided only qualitative explanations and lacked quantitative interpretability methods like SHAP for deeper insight into feature contributions.

Li et al. [12] proposed a CNN–BiLSTM–Attention model integrated with SHAP-based explainability for traffic accident severity prediction. The model effectively captured spatial–temporal dependencies while providing quantitative feature attributions through SHAP, identifying key contributors such as lighting conditions, time of day, and road surface type. This explainable hybrid approach improved prediction accuracy by over 10% compared to traditional deep models and enhanced transparency for data-driven traffic safety management.

Zhang et al. [13] developed an explainable hybrid CNN–BiLSTM–Attention–SHAP framework for traffic accident risk prediction using the UK STATS19 dataset. The model generated interpretable SHAP visualizations that highlighted the most influential factors affecting accident severity. By linking deep feature learning with explainable outputs, the study provided valuable insights for policy formulation and accident-prone zone identification, representing a significant advancement toward transparent and intelligent transportation systems.

Chen et al. [14] proposed a multimodal deep learning approach that integrates environmental, temporal, and traffic-related features to enable real-time accident forecasting. The framework effectively handled heterogeneous data sources and improved model adaptability to varying traffic conditions. However, it lacked an explainable AI component, making it difficult to interpret how individual factors influenced prediction outcomes, which limits its application in policy-driven traffic safety analysis.

Rashid et al. [15] addressed the issue of data imbalance in traffic accident datasets by applying SMOTE–Tomek resampling combined with feature normalization techniques. This preprocessing strategy enhanced the reliability and fairness of severity classification models by ensuring balanced class representation. While the approach improved model performance, it primarily focused on data preparation and did not incorporate explainable deep learning, limiting its interpretive potential for understanding key accident risk factors.

Gao et al. [16] integrated IoT-based real-time data streams with a Long Short-Term Memory (LSTM) deep learning network to predict traffic accidents dynamically. The system processed continuous sensor inputs such as vehicle speed, acceleration, and road occupancy to capture temporal dependencies and sudden traffic fluctuations. The model demonstrated high reliability in short-term forecasting and was capable of adapting to rapidly changing traffic conditions. However, despite its strong predictive accuracy, it did not incorporate explainability mechanisms such as SHAP or LIME, making it difficult to understand the contribution of individual IoT parameters to the prediction outcomes and reducing its interpretive utility for intelligent transportation management.

Xie et al. [17] emphasized the importance of explainable AI (XAI) in deep learning-based traffic accident prediction by integrating SHAP analysis into a CNN–BiLSTM framework. The proposed model not only improved prediction performance but also provided transparent interpretations of how various factors such as weather, lighting, and road surface conditions contributed to accident severity. The use of SHAP enhanced model trustworthiness and interpretability,

making the framework more suitable for deployment in real-world decision-support systems. This study demonstrated that coupling deep learning with explainability can significantly strengthen confidence in AI-driven transportation safety analytics.

Wang et al. [18] developed an explainable CNN–BiLSTM–Attention–SHAP framework using the UK STATS19 and US accident datasets. The study identified 15 critical features including weather conditions, road surface type, lighting, and temporal factors as major contributors to accident severity, establishing a new benchmark for interpretable deep learning in traffic risk prediction. By combining attention mechanisms with SHAP-based explainability, the framework not only enhanced predictive capability but also provided actionable insights for policymakers and traffic management authorities, enabling data-driven interventions to improve road safety.

From the reviewed studies, it is clear that traffic accident prediction has developed from traditional methods to advanced deep learning models. Earlier works used techniques such as Logistic Regression and Random Forest, which were easy to interpret but not suitable for handling complex spatial and temporal data. Later, deep learning models like CNN, LSTM, and BiLSTM were introduced to improve prediction accuracy by learning spatial and time-based patterns in traffic data. The addition of Attention Mechanisms helped these models focus on the most important factors such as weather, lighting, and road surface. Recently, Explainable AI (XAI) methods, especially SHAP, have been used to make these deep models more transparent and understandable. Overall, research has moved toward hybrid and interpretable models such as CNN–BiLSTM–Attention–SHAP, which combine both high accuracy and explainability, helping policymakers and engineers make better safety decisions.

3. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Traffic accident risk prediction and severity classification have traditionally relied on statistical and conventional machine learning methods applied to historical crash data. These approaches mainly depend on handcrafted features and predefined relationships among variables such as weather conditions, time of day, traffic flow, road type, and vehicle characteristics. Although these methods provide useful analytical insights, they often fail to capture complex spatial–temporal dependencies and nonlinear relationships present in real-world traffic environments.

Early machine learning techniques such as Logistic Regression (LR), Support Vector Machines (SVM), Decision Trees (DT), and Random Forests (RF) have been widely used for accident severity estimation. These models rely on manually engineered features such as road surface conditions, lighting, vehicle type, and environmental factors obtained through preprocessing. While these techniques achieve moderate prediction accuracy, their performance is limited by dependence on feature engineering, reduced generalization capability, and inability to effectively model temporal patterns in traffic data.

With the advancement of deep learning, models such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) have been explored to improve prediction accuracy through automatic feature extraction and temporal pattern learning. CNNs are effective in capturing spatial characteristics such as accident-prone road segments, while Recurrent Neural Networks, including Long Short-Term Memory (LSTM) and Bidirectional LSTM (BiLSTM) networks, can model sequential dependencies related to time, weather, and traffic flow. However, these deep learning models often function as black-box systems, offering limited interpretability and transparency.

Moreover, existing deep learning approaches generally require large, high-quality labeled datasets and involve high computational complexity. They also face challenges related to class imbalance, where severe accident cases are

underrepresented. Although hybrid CNN–LSTM models have shown improved predictive performance, most existing systems lack explainable artificial intelligence (XAI) integration. As a result, it becomes difficult for traffic authorities and policymakers to clearly understand the factors influencing accident risk and severity.

In addition, many existing traffic accident prediction systems are designed primarily for offline analysis and lack real-time adaptability. This limits their practical usefulness in dynamic traffic environments where conditions such as weather, traffic density, and road usage change rapidly. The absence of interpretability and real-time decision support further restricts the ability of these systems to assist traffic management authorities in implementing timely preventive measures and effective emergency response strategies.

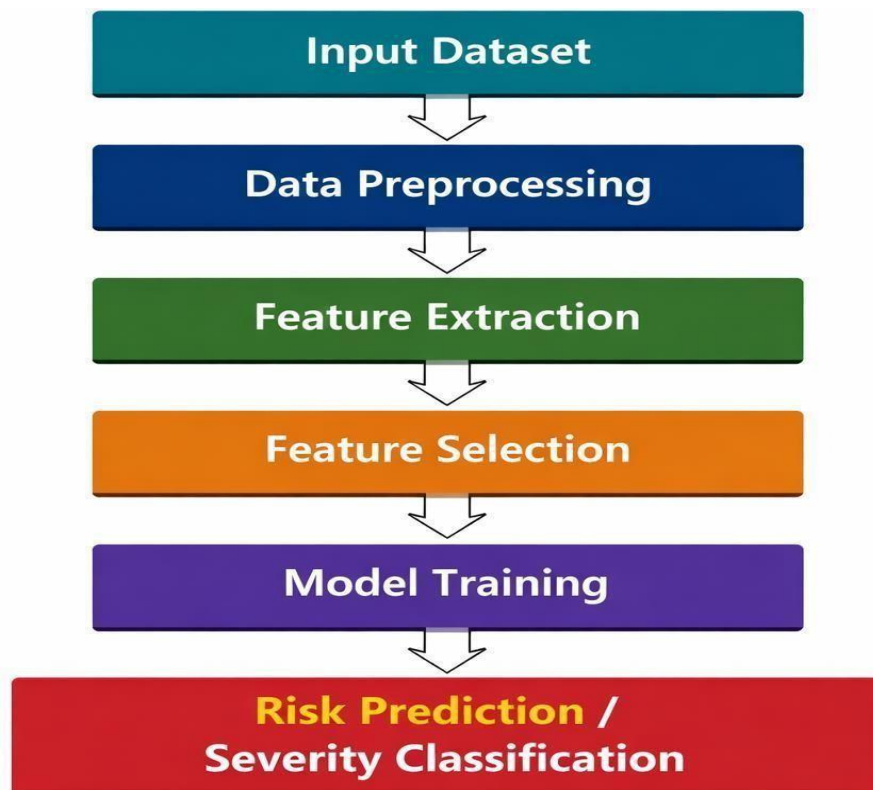


FIG 3.1. FLOW CHART OF EXISTING SYSTEM FOR TRAFFIC ACCIDENT RISK PREDICTION

This flowchart Fig. 3.1 illustrates the workflow followed in existing traffic accident risk prediction systems. The process begins with the input dataset, which includes various factors such as weather conditions, time of day, road type, vehicle details, and traffic characteristics. The dataset then undergoes data preprocessing to handle missing values, remove duplicate records, and normalize features to ensure consistency and suitability for further analysis.

After preprocessing, feature extraction is performed to identify meaningful variables that influence accident occurrence or severity. These features may include environmental factors, spatial attributes related to road geometry, and temporal patterns such as peak hours or seasonal variations. Feature selection techniques are subsequently applied to eliminate redundant or less significant features, thereby reducing computational complexity and improving model efficiency.

In the final stage, traditional machine learning algorithms such as Logistic Regression, Support Vector Machines, Decision Trees, or Random Forests are used to train the model and predict accident risk or severity levels. Although this existing system provides baseline predictions, it is limited in capturing complex spatial-temporal relationships and offers limited interpretability.

3.1.1 DISADVANTAGES OF THE EXISTING SYSTEM FOR TRAFFIC ACCIDENT RISK PREDICTION

Despite significant advancements in traffic accident risk prediction and severity classification, existing systems continue to face several limitations:

- **Limited Feature Understanding:**

Conventional models such as Logistic Regression, SVM, and Random Forest are unable to effectively capture complex spatial and temporal relationships among accident-related factors like weather, lighting, and road conditions.

- **Low Prediction Accuracy:**

These models rely on linear or shallow assumptions, making them less effective in handling nonlinear and dynamic traffic data patterns.

- **Lack of Explainability:** Many existing systems function as black-box models and do not provide clear explanations for their predictions, reducing trust and transparency in decision-making.

- **Data Imbalance Problem:**

Traffic accident datasets often contain a large number of minor accident cases and relatively few severe cases, leading to biased predictions and reduced performance for critical events.

- **Dependence on Static Data:**

Existing approaches primarily depend on historical or static datasets and lack adaptability to real-time changes such as traffic congestion, roadworks, or weather variations.

- **Poor Generalization:**

Models trained on specific regions or conditions may not generalize well to different locations or datasets due to variations in traffic behavior and infrastructure.

- **Computational Limitations:**

Some deep learning-based approaches require significant computational resources and training time, limiting their suitability for real-time prediction or deployment in low-resource environments.

3.2 PROPOSED SYSTEM

The proposed system introduces a hybrid deep learning framework combining Convolutional Neural Networks (CNN), Bidirectional Long Short-Term Memory (BiLSTM), Attention Mechanism, and SHAP (Shapley Additive Explanations) for traffic accident risk prediction and key factor identification.

This integrated approach leverages the spatial learning capability of CNN, the temporal sequence analysis of BiLSTM, the feature importance detection of Attention, and the interpretability provided by SHAP.

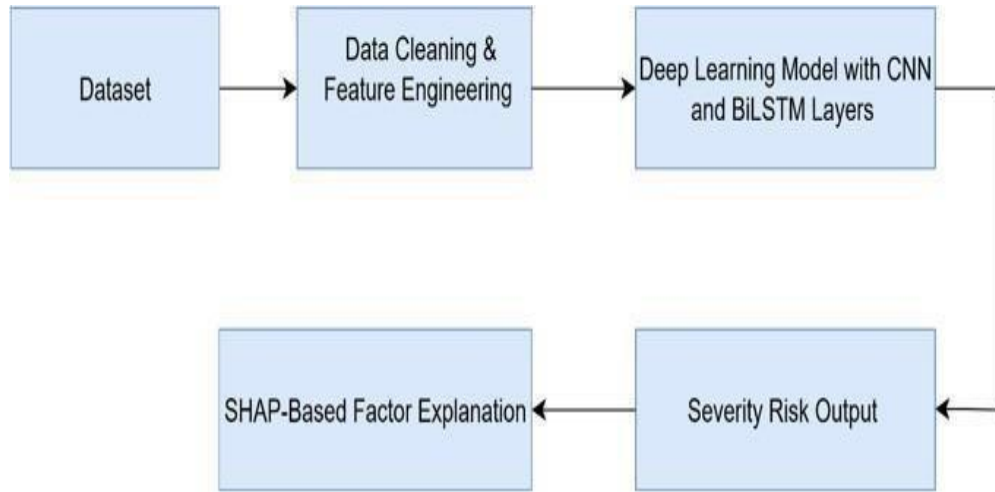


FIG 3.2. FLOW CHART OF PROPOSED SYSTEM

The workflow (Fig. 3.2) begins with data preprocessing, where the traffic accident dataset is cleaned, encoded, and normalized. Missing values are handled, categorical variables (such as weather, lighting, and road type) are encoded, and numerical features are scaled for consistency.

After preprocessing, the model performs feature extraction using CNN layers to capture spatial dependencies such as accident-prone locations and regional road layouts. These features are then passed to the BiLSTM layer, which learns temporal patterns such as time-of-day effects, seasonal variations, and traffic flow sequences.

The Attention mechanism follows, allowing the model to focus on the most influential risk factors (e.g., poor lighting, wet road surfaces, or high traffic density). This attention layer helps enhance the interpretability and predictive efficiency of the model by emphasizing critical inputs

Finally, SHAP explainability is applied to quantify the contribution of each feature toward the accident risk prediction. SHAP generates feature importance plots that identify the most impactful variables, supporting decision-making for transportation safety authorities.

Advantages over existing system:

1. **High Predictive Accuracy:**

The hybrid CNN–BiLSTM–Attention model captures both spatial and temporal dependencies, offering superior performance over traditional methods.

2. **Explainable Predictions:**

SHAP provides clear insights into which features most influence accident risk, increasing model transparency and trust.

3. **Improved Generalization:**

The combination of deep learning and explainability prevents overfitting and ensures robust predictions across multiple datasets and conditions.

4. **Real-Time Applicability:**

The model can process live data from IoT and traffic sensors for dynamic risk forecasting.

5. **Enhanced Decision Support:**

By identifying critical risk factors, the model assists policymakers and safety authorities in implementing effective road safety measures.

6. **Scalability and Adaptability:**

The framework can easily be extended to other regions, datasets, or transportation safety applications with minimal modifications.

3.3 FEASIBILITY STUDY

The proposed CNN–BiLSTM–Attention–SHAP framework for traffic accident risk prediction is designed to ensure high Prediction accuracy, scalability, and interpretability. The following feasibility analysis evaluates the system from technical, operational, and economic perspectives.

1. Technical Feasibility

- **Automated Feature Extraction:**

The CNN component automatically learns spatial patterns from traffic-related features such as road type, weather conditions, lighting, and location characteristics. This reduces dependency on manual feature engineering and enhances prediction accuracy.

- **Temporal Learning with BiLSTM:**

BiLSTM layers effectively capture temporal dependencies in accident data, including hourly, daily, and seasonal variations, enabling better modeling of time-dependent accident patterns.

- **Enhanced Interpretability with SHAP:**

The SHAP explainability module provides clear and quantitative insights into the contribution of each feature toward accident severity prediction, improving transparency and trustworthiness of the model.

- **Improved Accuracy and Generalization:**

The integration of CNN, BiLSTM, and Attention mechanisms allows the model to learn complex spatial–temporal relationships while reducing overfitting and improving generalization across dataset.

- **Scalability and Adaptability:**

The framework can be easily extended to different datasets (such as UK or US traffic accident data) and integrated with real-time IoT or traffic sensor data for continuous risk monitoring.

2. Operational Feasibility

- **Ease of Deployment:**

The system can be deployed on cloud platforms or integrated with existing traffic monitoring and decision-support systems used by transportation authorities.

- **User-Friendly Interface:**

The SHAP-based visual explanations help policymakers and analysts easily understand the key factors influencing accident risk and severity.

- **Data Handling:**

The framework supports large-scale and heterogeneous datasets containing environmental, temporal, vehicle, and road-related attributes.

- **Low Maintenance Requirements:**

Periodic retraining with updated accident data ensures sustained model performance without requiring major architectural changes.

3. Economic Feasibility

- **Cost-Effective Implementation:**

The model can be trained using publicly available datasets such as the **UK STATS19 dataset** and implemented using open-source deep learning frameworks, reducing development costs.

- **Efficient Resource Utilization:**

Model training and inference can be performed using standard GPU-enabled systems, reducing infrastructure and operational expenses.

- **Reduced Operational Costs:**

Automated accident risk prediction minimizes manual analysis efforts and improves response efficiency, leading to long-term cost savings for traffic authorities.

- **High Return on Investment:**

Improved accident risk prediction, enhanced safety planning, and data-driven policy decisions contribute to reduced accident-related economic losses.

3.4 USING COCOMO MODEL

The COCOMO (Constructive Cost Model) is a widely used software estimation technique for predicting development effort, time, and manpower required for a project. It provides an early estimation of project cost and schedule. For the proposed Traffic Accident Risk Prediction System using a CNN–BiLSTM–

Attention–SHAP framework, the Basic COCOMO Model is applied. Since the project involves moderate complexity—covering data preprocessing, deep learning model development, explainability integration, and web-based deployment—it is categorized as a Semi-Detached project.

The Basic COCOMO model relies on three key formulas for estimating effort, development time, and the number of people required:

- Effort (E) = $a \times (\text{KLOC})^b$ (measured in Person-Months)
- Development Time (T) = $c \times (E)^d$ (measured in Months)
- People Required (P) = E / T

In these formulas, KLOC refers to the estimated number of lines of code in thousands, and the constants (a , b , c , d) vary based on the project type. For Semi-Detached projects, the constants are $a = 3.0$, $b = 1.12$, $c = 2.5$, and $d = 0.35$.

Considering the entire project, including backend development, frontend design, and machine learning model implementation, the estimated size of the code is 10,000 lines of code, equivalent to 10 KLOC. Using the formulas, the estimated effort can be calculated as:

$$E = 3.0 \times (10)^{1.12} = 3.0 \times 13.18 \approx 39.54 \text{ Person-Months}$$

The development time is calculated as:

$$T = 2.5 \times (39.54)^{0.35} = 2.5 \times 4.23 \approx 10.58 \text{ Months}$$

Finally, the required number of people for the project is estimated using:

$$P = \frac{39.54}{10.58} \approx 3.74 \approx 4 \text{ People}$$

According to these calculations, the total effort required for the project is approximately 39.54 person-months, with an estimated development time of around 10.6 months. The project would ideally require a team of 4 members, which may include machine learning engineers, backend developers, frontend developers, and testers to ensure timely and efficient development. Several factors can influence these estimates, such as the complexity of the CNN–BiLSTM–Attention model, the size and quality of the UK and US traffic accident datasets, the integration of the Flask backend with the frontend interface, and the time allocated for testing, validation, and explainability analysis using SHAP. While the COCOMO model provides a reliable framework for initial estimation, real-world development may require adjustments based on challenges encountered during the project lifecycle.

4. SYSTEM REQUIREMENTS

4.1 SOFTWARE REQUIREMENTS

- | | |
|-------------------------|---------------------------------------|
| 1. Operating System | : Windows 11, 64-bit Operating System |
| 2. Hardware Accelerator | : CPU / GPU |
| 3. Coding Language | : Python |
| 4. Python distribution | : Google Colab Pro, Flask |
| 5. Browser | : AnyLatest Browser like Chrome |

4.2 REQUIREMENT ANALYSIS

The Traffic Accident Risk Prediction project aims to build an intelligent deep learning-based system that can accurately predict the severity of road accidents. The model combines Convolutional Neural Networks (CNNs) for spatial feature extraction, Bidirectional Long Short-Term Memory (BiLSTM) networks for temporal pattern learning, and an Attention mechanism to focus on key risk factors. To make the model more transparent, SHAP (Shapley Additive Explanations) is used to explain how each feature affects the prediction results.

The main functions of the system include loading traffic datasets, preprocessing data (such as normalization and encoding), training the CNN-BiLSTM-Attention model, and predicting accident severity. The system also displays model accuracy and SHAP graphs to show important factors like weather, lighting, road surface, and time of day that contribute to accidents.

In addition to core prediction functionality, the system supports explainable analysis to assist traffic authorities and researchers in understanding accident risk patterns. The SHAP-based explainability module generates visual representations such as feature importance plots, which clearly highlight the contribution of individual factors including weather conditions, road geometry, lighting, traffic density, and temporal attributes. This transparency enables informed decision-making and supports the development of targeted accident prevention strategies.

Non-functional requirements ensure the system is fast, reliable, accurate, and explainable. The model should handle large datasets efficiently using hardware with CPU or GPU support. The project requires Python 3.10 and deep learning libraries like TensorFlow or PyTorch. Deployment can be done locally or on a cloud platform, and users can access it through any modern web browser. The system is designed to be simple and user-friendly, helping traffic authorities and researchers analyze accident data, understand key risk factors, and improve road safety through data-driven insights.

4.3 HARDWARE REQUIREMENTS:

1. System Type : 64-bit operating system, x64-based processor
2. Cachelmemory : 4MB(Megabyte)
3. RAM : 16GB (gigabyte)
4. Hard Disk : 8GB
5. GPU : Intel® Iris® Xe Graphics

4.4 SOFTWARE

The Traffic Accident Risk Prediction system is developed using modern software tools and frameworks to ensure high performance, accuracy, and scalability. The project is implemented on Windows 11 (64-bit), providing a stable environment for deep learning and data processing tasks. The computations are performed using either CPU or GPU depending on availability, which ensures faster model training and prediction. The development language used is Python 3.10, chosen for its simplicity and wide support for machine learning and data science libraries. The model training and experimentation are conducted in Google Colab Pro, which offers optimized runtime with GPU acceleration for efficient deep learning model execution.

The project uses popular deep learning frameworks such as TensorFlow and PyTorch to implement the CNN–BiLSTM–Attention model. These frameworks handle feature extraction, sequence learning, and attention-based weighting for accurate accident risk prediction. The SHAP (Shapley Additive Explanations) library is integrated to provide interpretability, explaining how each feature (e.g., weather, lighting, road type) influences the model's predictions.

For data preprocessing and analysis, Pandas and NumPy are used to manage large traffic datasets and perform numerical computations efficiently. Scikit-learn supports data splitting, normalization, and evaluation metrics, while Matplotlib and Seaborn are used for visualization of results, such as accuracy plots and SHAP importance graphs.

The backend of the system is developed using Flask, enabling deployment as a web-based application. It allows users to upload data, run predictions, and view explainable outputs. The frontend uses HTML5, CSS3, and Bootstrap, ensuring a simple, responsive, and user-friendly interface that runs smoothly in any modern web browser like Google Chrome, Firefox, or Microsoft Edge.

Overall, this combination of software tools enables a powerful and transparent accident risk prediction framework that is accurate, explainable, and adaptable to different real-world traffic datasets.

4.5 SOFTWARE DESCRIPTION

The Traffic Accident Risk Prediction system requires a stable and updated operating environment to ensure smooth execution of deep learning and data analysis tasks. The project is designed to operate on Windows 11, 64-bit, which provides compatibility with the latest software tools, enhanced performance, and security features essential for research applications. The system primarily utilizes the CPU for model execution and backend operations. For faster training and large-scale computations, cloud-based environments such as Google Colab Pro are employed, providing access to GPU acceleration and optimized deep learning runtimes.

The development of the project is carried out using Python, which serves as the main programming language due to its flexibility and rich ecosystem of libraries for machine learning, data preprocessing, and visualization. Google Colab Pro is used for implementing and training the CNN-BiLSTM-Attention model, ensuring efficient runtime and easy collaboration. The backend of the system is built using Flask, a lightweight web framework that enables the deployment of the model as a web-based service, allowing users to input data, generate predictions, and visualize interpretability results interactively.

The SHAP library (Shapley Additive Explanations) is integrated to provide model transparency by identifying key factors influencing accident risk. Other essential Python libraries include TensorFlow/PyTorch for deep learning, scikit-learn for preprocessing and evaluation, Pandas and NumPy for data handling, and

Matplotlib/Seaborn for visual analysis. To access the system, any modern web browser such as Google Chrome, Mozilla Firefox, or Microsoft Edge can be used, ensuring a smooth and user-friendly experience.

5. SYSTEM DESIGN

5.1 SYSTEM ARCHITECTURE

This project focuses on predicting and analyzing traffic accident risk using an explainable deep learning framework. The proposed architecture combines Convolutional Neural Networks (CNN), Bidirectional Long Short-Term Memory (BiLSTM), and an Attention Mechanism, integrated with SHAP (Shapley Additive Explanations) for interpretability. The primary goal of this model is to accurately predict accident severity levels while identifying the most influential risk factors contributing to accident occurrences.

The model processes historical traffic accident data, such as that from the UK STATS19 dataset and US Accident Dataset (March 2023), which includes features like weather conditions, lighting, road surface, vehicle type, speed limit, and time of day. The architecture is designed to capture both spatial and temporal dependencies in these multi-dimensional datasets, ensuring precise and context-aware predictions. Initially, the dataset undergoes extensive preprocessing, including missing value imputation, normalization, and categorical encoding. Spatial features (e.g., road location, surface conditions) are learned through CNN layers, which extract local spatial patterns and environmental correlations. The output from CNN layers is then passed to BiLSTM networks, which effectively capture temporal dynamics such as time-of-day, seasonal variations, and accident frequency trends.

The Attention Mechanism plays a crucial role by assigning higher weights to the most relevant temporal features, allowing the model to focus on key risk indicators that significantly influence accident severity. This spatial-temporal learning process ensures that both environmental and time-dependent risk factors are accurately modeled.

For interpretability, the SHAP framework is employed after model training. SHAP provides feature-level explanations by quantifying each attribute's contribution to the prediction outcome. This helps identify the top influential features, such as weather, lighting, and road type, offering clear insights for policy-

makers and transportation authorities to develop effective safety interventions.

The model's performance is evaluated using multiple metrics, including Mean Absolute Error (MAE) for regression-based risk prediction, and Precision, Recall, and F1-score for classification-based severity analysis. Experimental results demonstrate that the CNN-BiLSTM-Attention-SHAP model outperforms traditional approaches like Logistic Regression, Random Forest, and standalone LSTM networks in both predictive accuracy and interpretability.

This architecture not only enhances predictive performance but also promotes transparency and explainability in decision-making. By integrating explainable AI techniques into traffic risk modeling, the system provides an effective tool for intelligent transportation systems, helping reduce accidents and improve overall road safety.

5.1.1 DataSet

The dataset utilized in this project is a curated collection of real-world road traffic accident records obtained from the UK STATS19 database and the US Fatality Analysis Reporting System (FARS). It forms the foundation for developing an advanced traffic accident risk prediction and severity classification framework. The dataset captures diverse accident scenarios, encompassing multiple influencing factors such as road geometry, weather, vehicle type, lighting conditions, time of day, and traffic flow. In addition to these, it also includes geospatial coordinates (latitude and longitude), enabling spatial mapping of accident-prone regions.

Each accident record in the dataset is labeled with a corresponding severity level Slight, Serious, or Fatal making it suitable for both regression-based risk analysis and classification-based severity prediction tasks. The dataset is systematically preprocessed to handle missing values, remove inconsistencies, and normalize numerical features. Categorical attributes such as weather, road type, and vehicle category are encoded using appropriate encoding techniques to ensure model compatibility.

Datasets link: <https://www.kaggle.com/datasets/tsiaras/uk-road-safety-accidents-and-vehicles> and <https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents>

To address the class imbalance problem, advanced balancing strategies such as

Random Oversampling and SMOTE (Synthetic Minority Oversampling Technique) are applied, ensuring fair representation of all severity categories. The dataset is then normalized and transformed into a structured format suitable for deep learning-based spatial-temporal modeling. This dataset plays a pivotal role in developing the CNN-BiLSTM-Attention-SHAP model. The CNN component extracts spatial correlations between accident features, while the BiLSTM captures temporal dependencies across time-series data such as hour, day, and season. The integrated Attention Mechanism enhances interpretability by focusing on the most significant risk factors, and the SHAP framework provides feature-level explanations to identify critical determinants like weather, lighting, and road conditions influencing accident severity. The dataset's comprehensive and heterogeneous nature makes it an invaluable resource for advancing research in intelligent transportation systems, enabling the development of explainable, data-driven tools for traffic accident risk prediction and prevention.

Feature	Details
Total Records	2.9 million (UK + US combined)
Geographic- al Coverage	UK (2011–2023), US (2016–2023)
Target Variable	Accident Severity (Slight / Serious / Fatal)
Key Attributes	Weather, Road Type, Lighting, Vehicle Type, Time, Speed, Traffic Flow
Spatial Features	Latitude, Longitude, Urban/Rural Area
Temporal Features	Hour, Day, Month, Season
Balancing Method	Balancing Method
Applications	Accident Severity Prediction and Risk Factor Analysis

TABLE 1 . DATASET DESCRIPTION

Severity Categories Included in the Dataset Slight Accidents:

Represent minor injuries and low-impact collisions. These cases form the majority of the dataset and exhibit diverse environmental and road conditions.

Serious Accidents:

Characterized by significant injuries requiring medical attention. These records typically reflect more complex accident patterns influenced by factors such as lighting, vehicle speed, and weather.

Fatal Accidents:

Represent high-impact and life-threatening collision events. Although fewer in number, these Records are critical for identifying high-risk conditions and evaluating model sensitivity to severe outcomes.

5.1.2 DATA PRE-PROCESSING

Data preprocessing is a critical phase in the development of the accident severity prediction framework, as it directly influences model performance, stability, and reliability. The raw accident records obtained from the UK STATS19 and US FARS databases contain heterogeneous formats, missing information, inconsistencies, and class imbalance issues. Therefore, a comprehensive multi-stage preprocessing pipeline was implemented. The following steps describe the full process:

Handling Missing Values: Missing numerical values are imputed using mean/median techniques, while missing categorical fields such as weather or lighting conditions are filled using the mode.

Removing Duplicates and Inconsistencies: Duplicate records, incorrect entries, and logically inconsistent accident reports are filtered out to maintain dataset integrity.

Encoding Categorical Variables: Text-based attributes such as road type, weather, vehicle category, and lighting conditions are converted into numerical form using label encoding and one-hot encoding.

Normalization of Numerical Features: Continuous variables (speed, traffic flow, time values) are standardized using methods such as Standard Scaler to ensure uniform scaling.

Outlier Detection and Correction: Outliers affecting model performance—such as unrealistic speed values—are identified and corrected or removed.

Spatial–Temporal Alignment: Location-based fields (latitude, longitude) and time-based attributes (hour, day, month, season) are formatted consistently for sequential modeling.

Class Imbalance Handling: To address the imbalance among Slight, Serious, and Fatal accident classes, oversampling and SMOTE techniques are applied. These preprocessing steps convert heterogeneous raw accident records into a clean, structured, and balanced dataset suitable for deep learning.

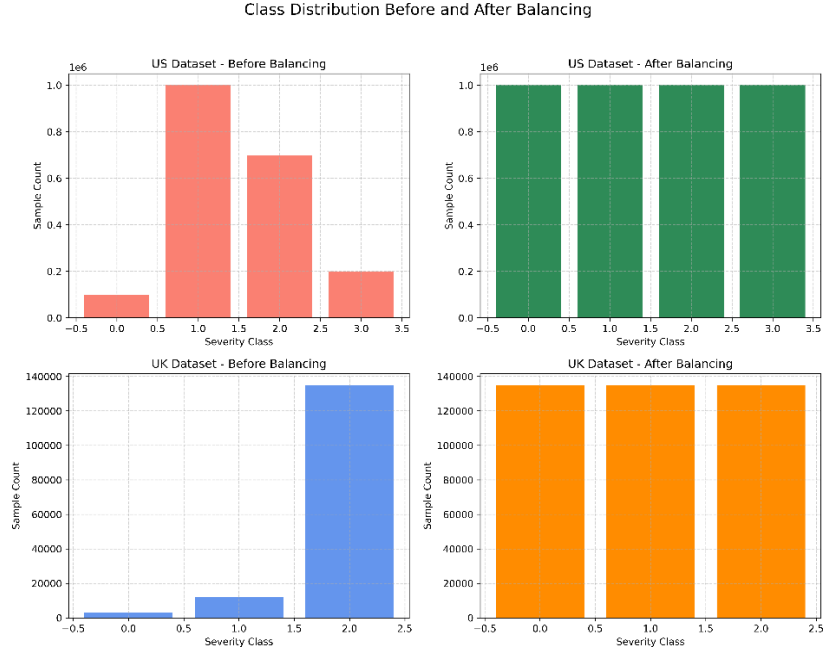


FIG: 5.1 IMAGE AFTER APPLYING THE PREPROCESSING TECHNIQUE.

5.1.3 FEATURE EXTRACTION

Feature Extraction is a crucial stage in the accident severity prediction framework. It involves selecting, transforming, and organizing the most relevant attributes from the preprocessed dataset to improve model performance and interpretability. Since accident data is highly heterogeneous, the extracted features include spatial, temporal, environmental, vehicle-specific, and road-related attributes. These features enable the CNN–BiLSTM–Attention architecture to learn both spatial correlations and temporal patterns effectively.

Spatial Features: Latitude, longitude, region type (urban/rural), and road layout help in identifying accident-prone geographical areas.

Temporal Features: Hour of the day, day of the week, month, and season capture time-dependent accident patterns.

Environmental Features: Weather condition, lighting environment, road surface condition, and visibility factors represent external influences on accident severity.

Vehicle and Traffic Features: Vehicle type, traffic flow, speed, and number of vehicles involved describe dynamic road conditions.

Road Characteristics: Road type, junction layout, curvature, and road geometry support modeling of structural risk factors.

These extracted features collectively support the CNN for spatial learning, the BiLSTM for temporal trend identification, and the Attention mechanism for identifying influential accident factors.

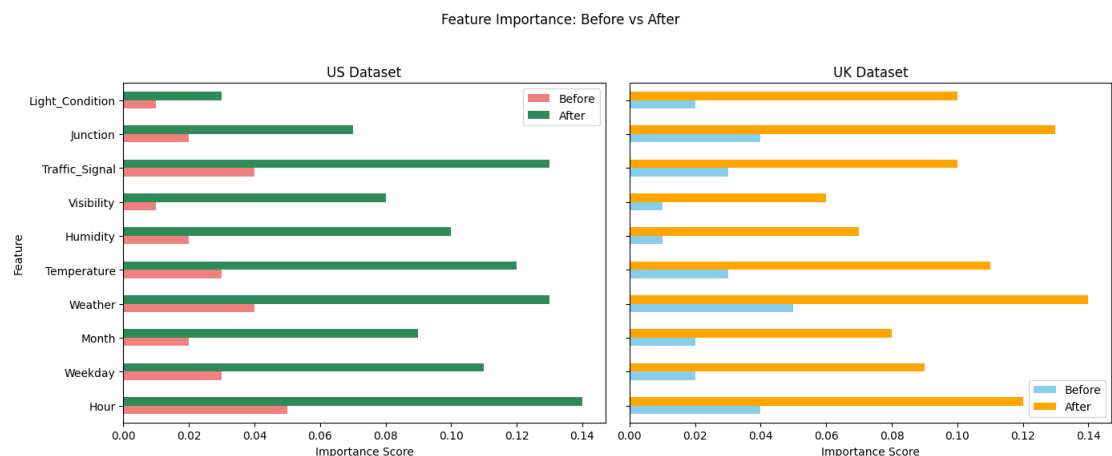


FIG: 5.2 FEATURE IMPORTANCE BEFORE vs AFTER

5.1.4 MODEL BUILDING:

Model building in the context of Deep Learning refers to the systematic process of designing, developing, and integrating multiple neural network components to accurately learn patterns from complex and heterogeneous data. For traffic accident severity prediction, this process involves constructing a hybrid model capable of capturing both spatial correlations and temporal dependencies present in real-world accident records. In this project, the proposed architecture combines Convolutional Neural Networks (CNNs), Bidirectional Long Short-Term Memory (BiLSTM) networks, and an Attention Mechanism, each contributing uniquely to the overall performance, robustness, and interpretability of the system. The model is designed to process structured accident-related features—including weather conditions, lighting, traffic flow, road geometry, environmental factors, and time-based variables—and Produce

accurate severity predictions across categories such as Slight, Serious, and Fatal.

Convolutional Neural Networks:

Convolutional Neural Networks are utilized in this system to extract spatial feature representations from the input data. Although CNNs are traditionally applied to image data, 1D convolutional layers are highly effective for structured numerical datasets as well, especially when the features exhibit local relationships. In accident data, variables such as weather–visibility, road type–lighting, or speed–traffic flow often interact in meaningful ways. The CNN learns these localized patterns by applying learnable convolutional kernels that slide across the input feature vector and highlight significant correlations. The resulting feature maps emphasize important accident-related characteristics while reducing noise and redundancy. Through convolution and non-linear activation, the CNN forms the foundational layer of the model by transforming raw features into richer spatial representations that support deeper learning in subsequent layers.

Layers in Convolutional Neural Networks:

Convolutional Neural Networks (CNNs) are widely used for automatic feature extraction due to their ability to learn spatial patterns from input data. In this project, CNN layers are employed to extract meaningful spatial features from traffic accident data.

- **Input Layer** - The input layer receives the preprocessed traffic accident data in the form of numerical feature vectors representing factors such as weather conditions, road type, lighting, and traffic characteristics.
- **Convolutional Layer** - The convolutional layer applies learnable filters (kernels) to the input data to extract important spatial patterns. In the proposed system, this layer helps identify complex relationships among accident-related features, enabling the model to capture underlying risk patterns.
- **Activation Layer** - Activation functions introduce non-linearity into the network, allowing it to model complex relationships. The Rectified Linear Unit (ReLU) activation function is commonly used due to its computational efficiency and ability to mitigate vanishing gradient problems. Other activation functions such as Sigmoid and Tanh may also be used depending on the task.
- **Pooling Layer** - The pooling layer reduces the dimensionality of feature maps while retaining essential information. This reduces computational complexity and helps prevent overfitting. Common pooling techniques include:

Max Pooling: Retains the maximum value from each pooling window.

Average Pooling: Computes the average value within each pooling window.

- **Flattening Layer** - The flattening layer converts the multi-dimensional feature maps into a one-dimensional vector, making it suitable for input into fully connected layers.
- **Fully Connected Layer (Dense Layer)** - The fully connected layer combines the extracted features to learn higher-level representations. Each neuron in this layer is connected to all neurons in the previous layer, enabling effective classification or regression of accident severity.
- **Output Layer** - Produces the final predictions of the network. Number of neurons matches the number of target classes.

Common activation functions:

SoftMax: For multi-class classification, outputs probabilities for each class.

Sigmoid: For binary classification, outputs a probability between 0 and 1.

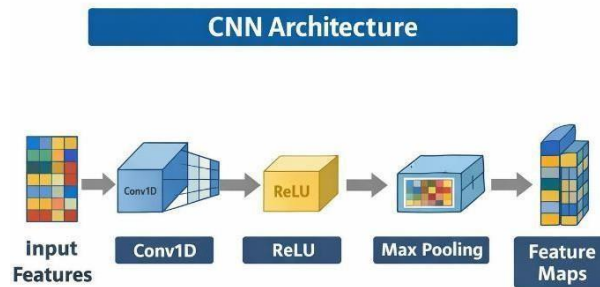


FIG 5.3 CNN MODEL ARCHITECTURE

Bidirectional Long Short-Term Memory (BiLSTM):

After spatial features are extracted using the Convolutional Neural Network (CNN), the model employs a Bidirectional Long Short-Term Memory (BiLSTM) layer to learn temporal and sequential dependencies within traffic accident data. Accident severity is strongly influenced by time-dependent patterns; for example, specific hours of the day, weekends, seasonal variations, and changing weather conditions can significantly impact accident outcomes.

Conventional neural networks and standard Recurrent Neural Networks (RNNs) struggle to capture long-term dependencies due to issues such as vanishing gradients. BiLSTM networks overcome these limitations by incorporating memory

cells and gating mechanisms that effectively retain relevant information over extended time intervals.

A BiLSTM processes the input sequence in both forward and backward directions, enabling the model to consider past and future contextual information simultaneously. This bidirectional learning allows the model to understand how earlier factors (such as visibility or lighting conditions) and later factors (such as road surface or traffic density) jointly influence accident severity. As a result, the model gains a more comprehensive temporal representation of accident patterns.

By modeling traffic accident data as a sequence and analyzing it holistically, the BiLSTM layer enhances the system’s ability to recognize meaningful temporal trends and dependencies. This significantly improves prediction accuracy, stability, and robustness, making BiLSTM a critical component of the proposed CNN–BiLSTM–Attention framework.

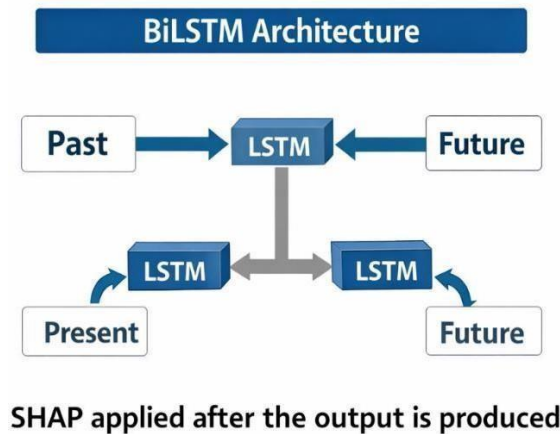


FIG 5.4 BIDIRECTIONAL LONG SHORT-TERM MEMORY ARCHITECTURE

Attention Mechanism:

After learning spatial features through Convolutional Neural Networks (CNN) and capturing temporal dependencies using Bidirectional Long Short-Term Memory (BiLSTM), an Attention Mechanism is applied to further enhance the prediction performance of the Traffic Accident Risk Prediction system.

In complex accident datasets, not all features or time steps contribute equally to accident severity. Certain conditions such as poor weather, low visibility, peak traffic hours, or dangerous road geometry may play a more significant role than others. The attention mechanism enables the model to automatically identify and focus on the most important features and time steps, while assigning lower importance to less relevant information.

The attention layer works by computing attention weights for the BiLSTM output sequence. These weights indicate the relative importance of each temporal feature representation. By multiplying the BiLSTM outputs with their corresponding attention weights and aggregating them, the model generates a context vector that highlights critical accident-related patterns.

CNN–BiLSTM–Attention Hybrid Architecture:

The CNN–BiLSTM–Attention Hybrid Architecture integrates spatial feature learning, temporal dependency modeling, and adaptive feature prioritization into a unified deep learning framework for accurate and explainable traffic accident severity prediction.

The architecture begins with a Convolutional Neural Network (CNN) that processes the preprocessed traffic accident data. Using 1D convolutional layers, the CNN extracts spatial feature representations, capturing localized relationships among variables such as weather conditions, road type, lighting, traffic density, and vehicle characteristics. Convolution and activation layers transform raw inputs into high-level spatial feature maps while reducing noise and redundancy.

The extracted spatial features are then passed to a Bidirectional Long Short-Term Memory (BiLSTM) layer. The BiLSTM models temporal dependencies by analyzing the feature sequences in both forward and backward directions. This allows the model to understand how past and future contextual factors—such as time of day, seasonal patterns, and evolving traffic conditions—jointly influence accident severity.

Following the BiLSTM, an Attention Mechanism is applied to dynamically assign importance weights to the temporal feature representations. The attention layer identifies the most critical time steps and feature combinations that contribute

to accident severity, generating a context vector that emphasizes high-risk patterns while suppressing less relevant information.

Finally, the attention-weighted context vector is fed into fully connected layers for risk prediction or severity classification. This hybrid architecture effectively combines spatial–temporal learning with interpretability, resulting in improved prediction accuracy, reduced overfitting, and better generalization across multiple datasets such as UK and US traffic accident records.

This CNN–BiLSTM–Attention framework forms the core predictive engine of the proposed explainable traffic accident risk prediction system.

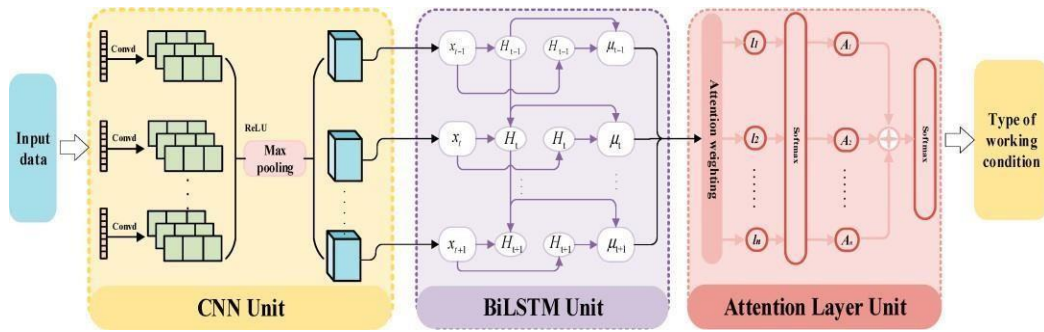


FIG 5.5 CNN–BiLSTM–ATTENTION HYBRID ARCHITECTURE

SHAP Explainability Module:

To ensure that the proposed CNN–BiLSTM–Attention model is not only accurate but also transparent and trustworthy, a SHAP (SHapley Additive exPlanations)–based explainability module is integrated into the system. Although deep learning models are highly effective in capturing complex spatial–temporal patterns, they are often criticized for operating as black boxes, making it difficult to understand the reasoning behind their predictions. SHAP addresses this limitation by providing a theoretically grounded and interpretable explanation of model outputs.

SHAP is based on Shapley values from cooperative game theory, which quantify the contribution of each input feature to the final prediction. In the context of traffic accident risk prediction, SHAP evaluates how features such as weather conditions, lighting, road surface, traffic density, speed, junction type, and time of day influence the predicted accident severity. A positive SHAP value indicates that a feature increases the likelihood of a severe accident, whereas a negative value suggests a mitigating effect.

The SHAP module is applied after the model generates its prediction, enabling both local and global interpretability. Local explanations provide instance-level insights, showing why a particular accident case was classified as severe or non-severe. Global explanations summarize feature importance across the entire dataset, identifying the most influential risk factors common to multiple accident scenarios.

By integrating SHAP, the proposed framework allows traffic authorities, policymakers, and researchers to clearly understand why a prediction is made, not just what the prediction is. This transparency supports informed decision-making, targeted safety interventions, and increased trust in intelligent transportation systems.

Advantages of SHAP Integration:

- Improves transparency of deep learning predictions
- Identifies key accident risk factors clearly
- Supports both global and case-specific analysis
- Enhances trust and reliability for real-world deployment
- Assists policymakers in data-driven road safety planning

5.1.5 CLASSIFICATION

Classification using Proposed CNN–BiLSTM–Attention Hybrid Model:

The proposed CNN–BiLSTM–Attention hybrid architecture is specifically designed to capture both spatial and temporal dependencies within traffic accident datasets. Unlike traditional machine learning models that rely heavily on manually engineered features, this hybrid deep learning approach automatically learns complex relationships among accident-related variables such as weather, lighting conditions, road type, traffic flow, and time.

The classification process begins with the Convolutional Neural Network (CNN) component, which processes structured accident features to extract high-level spatial patterns. CNN layers act as feature detectors that identify correlations among parameters—for example, how combinations of weather conditions and road characteristics influence accident risk. Activation functions such as ReLU introduce non-linearity, enabling the model to learn complex interactions, while pooling layers reduce dimensionality and control overfitting by retaining the most significant information.

The extracted spatial features are then passed to the Bidirectional Long Short-Term Memory (BiLSTM) layer, which captures temporal dependencies present in accident data. Accident severity is strongly influenced by time-based patterns such as hourly variations, weekday–weekend differences, and seasonal trends. By processing the sequence in both forward and backward directions, the BiLSTM enables the model to understand how past and future contextual factors jointly influence accident outcomes, making it highly suitable for traffic-related time-series data.

To further enhance prediction accuracy and interpretability, an Attention Mechanism is integrated after the BiLSTM layer. While CNN and BiLSTM capture spatial and temporal features respectively, not all features contribute equally to the final prediction. The Attention layer assigns higher weights to more influential factors—such as poor lighting, adverse weather, or high-speed road segments—allowing the model to focus on the most critical risk indicators. This selective weighting improves both model precision and transparency.

Finally, the attention-weighted feature representation is passed through fully connected layers to produce the final output, classifying each accident instance into severity categories such as Slight, Serious, or Fatal. The model is trained using cross-entropy loss, and optimization algorithms such as Adam are employed to ensure efficient convergence during training.

Compared to traditional models such as Logistic Regression or Random Forests, the CNN–BiLSTM–Attention architecture provides a more robust, scalable, and accurate solution for traffic accident severity prediction. CNN captures spatial relationships, BiLSTM models temporal dependencies, and the Attention mechanism enhances interpretability by highlighting key contributing factors.

To further improve transparency, SHAP (SHapley Additive exPlanations) is applied during post-classification analysis. SHAP quantifies the contribution of each input feature to the final prediction, offering clear insights into the most influential accident risk factors. For example, SHAP can reveal how conditions such as rainy weather, night-time driving, or curved road geometry affect severity outcomes, making the model trustworthy and suitable for real-world deployment.

During inference, accident data flows sequentially through the CNN for spatial feature extraction, the BiLSTM for temporal learning, and the Attention layer for weighted decision-making, resulting in an accurate and explainable severity classification. This hybrid approach enhances reliability and reduces false predictions, making it well-suited for intelligent transportation systems, emergency response planning, and real-time traffic risk monitoring.

5.2 MODULES

In the context of software development, a module is a self-contained, independent unit of code that performs a specific task or functionality within a larger system.

CNN-BiLSTM-Attention Accident Prediction Project Modules:

- 1. Data Collection Module:** Collects accident data from UK and US datasets and loads them into the system.

Sample Code:

```
import pandas as pd
def load_datasets(uk_accident_path, uk_vehicle_path, us_dataset_path):
    # Load UK accident and vehicle datasets
    uk_accident = pd.read_csv(uk_accident_path, low_memory=False)
    uk_vehicle = pd.read_csv(uk_vehicle_path, low_memory=False)
    # Merge UK datasets using Accident_Index
    uk_data = pd.merge(
        uk_accident,
        uk_vehicle,
        on='Accident_Index',
        how='inner'
    )
    # Load US accident dataset
    us_data = pd.read_csv(us_dataset_path)
    # Combine UK and US datasets
    data = pd.concat([uk_data, us_data], ignore_index=True)
    return data
```

- 2. Preprocessing Module:** Cleans the data by removing missing values, converting time features, encoding categories, and normalizing numerical values.

Sample Code:

```
from sklearn.preprocessing import LabelEncoder,
StandardScaler
def preprocess_data(df):
    df = df.fillna(df.mean())

    encoder = LabelEncoder()
```

```

df['Weather'] = encoder.fit_transform(df['Weather'])

scaler = StandardScaler()

df[['Speed','Traffic_Flow']] =

scaler.fit_transform(df[['Speed','Traffic_Flow']])

return df

```

- 3. Data Balancing Module (SMOTE):** Resolves class imbalance using SMOTE / Random Oversampling.

Sample Code:

```

from imblearn.over_sampling import SMOTE

def balance_data(X, y):

sm = SMOTE()

X_res, y_res = sm.fit_resample(X, y)

return X_res, y_res

```

- 4. Feature Engineering Module:** Generates temporal features and structures input for CNN.

Sample Code:

```

df_uk['DayOfWeek'] = df_uk['Date'].dt.dayofweek

df_us['Month'] = df_us['Start_Time'].dt.month

```

- 5. CNN Feature Extraction Module:** Extracts spatial feature patterns from structured accident data.

Sample Code:

```

import torch.nn as nn

class CNN_Module(nn.Module):

def _init_(self):

    super()._init_()

    self.conv = nn.Conv1d(1, 64, kernel_size=3)

    self.relu = nn.ReLU()

    self.pool = nn.MaxPool1d(2)

def forward(self, x):

    x = self.pool(self.relu(self.conv(x)))

return x

```

6. **BiLSTM Temporal Learning Module:** Learns accident patterns across time.

Sample Code:

```
class BiLSTM_Module(nn.Module):
    def __init__(self):
        super().__init__()
        self.lstm = nn.LSTM(64, 128, batch_first=True, bidirectional=True)
    def forward(self, x):
        output, _ = self.lstm(x)
    return output
```

7. **Attention Mechanism Module:** Assigns weights to important features to improve interpretability.

Sample Code:

```
class Attention(nn.Module):
    def __init__(self, features): super().__init__()
    self.att = nn.Linear(features, 1)
    def forward(self, x):
        weights = torch.softmax(self.att(x), dim=1)
        return (weights * x).sum(dim=1)
```

8. **Classification Module:** Final Dense layers for predicting accident severity: Slight, Serious, Fatal.

Sample Code:

```
class Classifier(nn.Module): def __init__(self):
    super().__init__()
    self.fc = nn.Linear(256, 3)
    def forward(self, x):
        return nn.Softmax(dim=1)(self.fc(x))
```

9. **SHAP Module:** Generates feature importance and model explanation

Sample Code:

```
import shap
def explain_model(model, sample):
    explainer = shap.DeepExplainer(model,
    sample)
    return explainer.shap_values(sample)
```

10. **Evaluation Module:** Evaluates model performance.

Sample Code:

```
from sklearn.metrics import
accuracy_score def
evaluate_model(true_labels, predictions):
    return accuracy_score(true_labels, predictions)
```

11. **Flask Backend Module:** Manages API endpoints.

Sample Code:

```
from flask import Flask, request, jsonify
```

```

app      =      Flask(__name__)
@app.route('/predict', methods=['POST'])
def predict():
    file = request.files['file']
    # Process file and return prediction return
    jsonify({'result': 'Prediction Here'})
if __name__ == '__main__': app.run(debug=True)

```

12. **Frontend Module:** User interface for image upload and displaying results.

Sample Code:

```

<form action="/predict" method="post"
enctype="multipart/form-data">
<input type="file" name="file">
<input type="submit" value="Upload">
</form>

```

13. **File Management Module:** Manages file storage and cleanup.

Sample Code:

```

import os
def delete_file(file_path):
if os.path.exists(file_path):
os.remove(file_path)

```

5.3 UML DIAGRAMS

The workflow of the CNN–BiLSTM–Attention model for road accident severity prediction begins with collecting and preprocessing accident data from the UK STATS19 and US FARS datasets. Preprocessing includes essential steps such as handling missing values, removing outliers, encoding categorical features (weather, lighting, road type), normalizing numerical attributes (speed, traffic flow), and balancing the classes using SMOTE to ensure equal representation of severity categories (Slight, Serious, Fatal).

Feature engineering is then performed to extract and derive meaningful attributes such as hour, day, month, and season of the accident, along with spatial features like latitude, longitude, and urban/rural indicators. The data is then transformed into structured feature matrices suitable for CNN-based spatial feature extraction.

The CNN–BiLSTM–Attention hybrid model is trained on the preprocessed data. The CNN module extracts spatial correlations among accident attributes such as weather and lighting, while the BiLSTM learns temporal dependencies across time patterns. The integrated Attention mechanism highlights the most influential features contributing to accident severity prediction. After training, the model's performance is evaluated using metrics such as accuracy, MAE, precision, recall, F1-score, and confusion matrix to assess predictive effectiveness.

Once the hybrid model is successfully trained, it is saved for deployment and future usage. Saving the trained model preserves its layers, learned weights, and parameters, allowing the system to perform predictions without retraining. This is particularly important for real-time Intelligent Transportation Systems (ITS), where predictions must be generated rapidly and reliably. Following evaluation, a performance report is generated summarizing the model's accuracy and the SHAP-based feature interpretation results. These insights help identify key accident contributors such as weather, lighting, traffic flow, and road type. In real-world applications, explainability is crucial because it builds trust among traffic analysts, policy makers, and road safety authorities by providing transparent justification for each prediction.

This flowchart (Fig 5.7) outlines the complete process of building and evaluating the accident severity prediction pipeline.

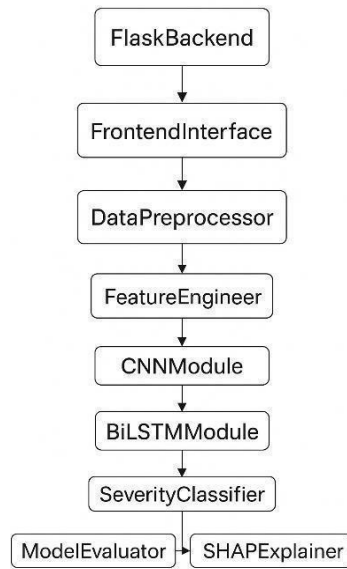


FIG 5.6 DESIGN OVERVIEW

In The system begins with accident data acquisition from official sources, followed by data preprocessing tasks such as balancing, encoding, and normalization. The processed data is then passed to the CNN–BiLSTM–Attention model. CNN extracts spatial patterns; BiLSTM captures temporal sequences; Attention emphasizes critical features. After training, the model is saved and evaluated using performance metrics and SHAP explainability.

In the CNN–BiLSTM–Attention accident prediction project, UML (Unified Modeling Language) diagrams play an essential role in visualizing the system’s architecture and workflow.

Class Diagram Overview

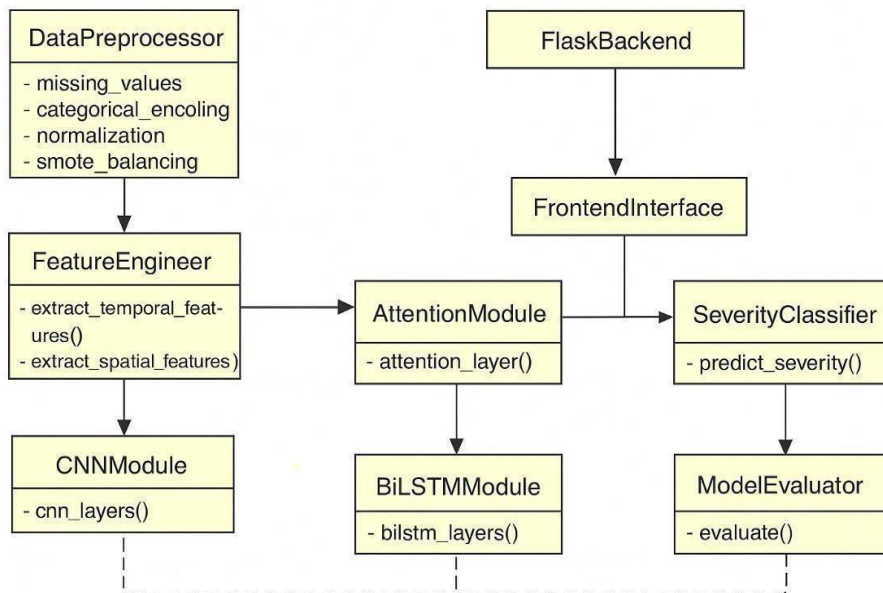
The Data Preprocessor class handles tasks such as loading accident data, cleaning it, performing encoding, normalization, and applying SMOTE for balancing. The Feature Engineer class derives new features such as hour, day, season, and traffic indicators.

The CNN Module extracts spatial feature maps, while the BiLSTM Module learns temporal patterns. The Attention Module highlights important factors influencing severity. The Severity Classifier performs final prediction of Slight, Serious, or Fatal accidents. The Model Evaluator class computes performance metrics including accuracy, MAE, precision, recall, F1-score, and confusion matrix. The SHAP Explainer generates interpretable feature importance visuals. The Flask Backend manages API endpoints for prediction requests, while the User Interface allows users to input accident data and view results.

SequenceDiagram Overview

The sequence begins when the user submits accident details using the frontend form. The Flask Backend receives the request and forwards it to the Data Preprocessor, which cleans and structures the data. The Feature Engineer derives additional attributes and passes them to the CNN Module, which extracts spatial features. The output is then processed by the BiLSTM Module, followed by the Attention Module, which assigns feature weights. The Severity Classifier predicts the accident severity. Once the prediction is generated, the SHAP Explainer produces interpretability outputs, which are included in the result. The prediction and explanation are returned to the frontend for display.

Overall, UML diagrams help visualize the roles of various components, simplify system understanding, and improve maintainability of the accident severity predictions system.



The Fig 5.7 UML Class Diagram of the Proposed CNN–BiLSTM–Attention

The UML diagram (Fig 5.8) illustrates the modular architecture and data flow of the CNN– BiLSTM–Attention model. The Flask Backend handles API operations such as receiving accident data, coordinating preprocessing, model execution, and returning prediction results. The Frontend Interface allows users to provide accident information and view predictions with SHAP- based explanations. The Data Preprocessor performs initial tasks such as imputing missing values, encoding categorical data, normalization, and SMOTE-based class balancing. The Feature Engineer extracts temporal and spatial features. The CNN Module processes structured inputs to learn spatial correlations. The BiLSTM Module models sequential temporal patterns. The Attention Module calculates attention weights to focus on influential factors. These processed features are passed to the Severity Classifier, which performs the final classification of accident severity. The Model Evaluator computes various performance metrics, while the SHAPExplainer generates feature importance reports for interpretability. The results are then returned to users, making the system reliable, explainable, and suitable for intelligent transportation applications.

6. IMPLEMENTATION

6.1 MODEL IMPLEMENTATION

CNN-BiLSTM-Attention Model :

```
# Import required libraries
import pandas as pd
import numpy as np
import torch
import torch.nn as nn
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from imblearn.combine import SMOTETomek
from sklearn.metrics import accuracy_score,
classification_report, mean_absolute_error

# Load and merge UK and US datasets
def load_datasets(uk_accident_path, uk_vehicle_path, us_dataset_path):
    uk_accident = pd.read_csv(uk_accident_path, low_memory=False)
    uk_vehicle = pd.read_csv(uk_vehicle_path, low_memory=False)
    uk_data = pd.merge(uk_accident, uk_vehicle,
on='Accident_Index', how='inner')
    us_data = pd.read_csv(us_dataset_path, low_memory=False)
    data = pd.concat([uk_data, us_data], ignore_index=True)
    return data

# Handle missing values
def preprocess_data(df):
    df = df.dropna(thresh=int(0.9 * len(df.columns)))
    df = df.fillna(method='ffill')

# Encode categorical features
for col in df.columns:
    if df[col].dtype == 'object':
        df[col] = LabelEncoder().fit_transform(df[col].astype(str))
    return df

# Split features and target
def split_and_scale(df, target_col='Accident_Severity'):
    X = df.drop(target_col, axis=1)
    y = df[target_col]

# Balance classes
X_bal, y_bal = SMOTETomek(random_state=42).fit_resample(X, y)
```

```

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_bal)
return X_scaled, y_bal

# Temporal feature extraction
def feature_engineering(df):
    if 'Date' in df.columns:
        df['Date'] = pd.to_datetime(df['Date'], errors='coerce')
        df['Hour'] = df['Date'].dt.hour
        df['Day'] = df['Date'].dt.day
        df['Month'] = df['Date'].dt.month
        df['Season'] = (df['Month'] % 12 + 3) // 3
        df_uk['Year'] = df_uk['Date'].dt.year
        df['DayOfWeek'] = df['Date'].dt.dayofweek
    df = df.drop(['Date', 'Time'], axis=1, errors='ignore')
    return df

# CNN-BiLSTM-Attention Model
class Attention(nn.Module):
    def __init__(self, hidden_dim):
        super().__init__()
        self.attn = nn.Linear(hidden_dim * 2, 1)
    def forward(self, lstm_out):
        weights = torch.softmax(self.attn(lstm_out), dim=1)
        context = torch.sum(weights * lstm_out, dim=1)
        return context

class CNN_BiLSTM_Attention(nn.Module):
    def __init__(self, input_dim, hidden_dim=64, num_classes=3):
        super().__init__()
        self.cnn = nn.Sequential(
            nn.Conv1d(1, 32, kernel_size=3, padding=1),
            nn.ReLU(),
            nn.MaxPool1d(2),
            nn.Conv1d(32, 64, kernel_size=3, padding=1),
            nn.ReLU(),
            nn.MaxPool1d(2)
        )
        self.bilstm = nn.LSTM(
            input_size=64,
            hidden_size=hidden_dim,
            batch_first=True,
            bidirectional=True

```

```

)
self.attention = Attention(hidden_dim)
self.fc = nn.Linear(hidden_dim * 2, num_classes)

def forward (self, x):
    x = x.unsqueeze(1)
    x = self.cnn(x)
    x = x.permute(0, 2, 1)
    lstm_out, _ = self.bilstm(x)
    attn_out = self.attention(lstm_out)
    return self.fc(attn_out)

# Model Training and Evaluation
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y_bal, test_size=0.2, random_state=42)
X_train = torch.tensor(X_train, dtype=torch.float32)
X_test = torch.tensor(X_test, dtype=torch.float32)
y_train = torch.tensor(y_train.values, dtype=torch.long)
y_test = torch.tensor(y_test.values, dtype=torch.long)
model = CNN_BiLSTM_Attention(input_dim=X_train.shape[1])
criterion = nn.CrossEntropyLoss()
optimizer = torch.optim.Adam(model.parameters(), lr=0.001)
for epoch in range(20):
    model.train()
    optimizer.zero_grad()
    outputs = model(X_train)
    loss = criterion(outputs, y_train)
    loss.backward()
    optimizer.step()
    if (epoch + 1) % 5 == 0:
        print(f'Epoch {epoch+1}, Loss: {loss.item():.4f}')
    model.eval()
    with torch.no_grad():
        preds = model(X_test)
        y_pred = torch.argmax(preds, dim=1)
        print("Accuracy:", accuracy_score(y_test, y_pred))
        print(classification_report(y_test, y_pred))

#SHAP
import shap
explainer = shap.DeepExplainer(model, X_train[:100])
shap_values = explainer.shap_values(X_test[:50])

```

```

shap.summary_plot(shap_values, X_test[:50].numpy())

#Confusion Matrix
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix
# Compute confusion matrix
cm= confusion_matrix(y_true, y_pred)
# Define class labels
class_names = ["Slight", "Severe"]
# Plot confusion matrix
plt.figure(figsize=(6,5))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues", cbar=True,
            xticklabels=class_names, yticklabels=class_names)
plt.title("Confusion Matrix")
plt.xlabel("Predicted label")
plt.ylabel("True label")
plt.show()

#Logistic Regression
from sklearn.linear_model import LogisticRegression
# Train Logistic Regression on same preprocessed data
lr_model = LogisticRegression(max_iter=1000)
lr_model.fit(X_train, y_train_mapped)
y_lr_pred = lr_model.predict(X_test)
# Evaluation
print("📊 Logistic Regression Classification Report:")
print(classification_report(y_test_mapped, y_lr_pred, zero_division=0))
# Accuracy
acc_lr = accuracy_score(y_test_mapped, y_lr_pred)
print(f"✅ Logistic Regression Accuracy: {acc_lr:.4f}")

# Confusion Matrix
cm_lr = confusion_matrix(y_test_mapped, y_lr_pred)
disp_lr = ConfusionMatrixDisplay(confusion_matrix=cm_lr,
display_labels=['Slight', 'Serious']) # Modified labels here
disp_lr.plot(cmap=plt.cm.Greens)
plt.title("Logistic Regression - Confusion Matrix")
plt.show()

import matplotlib.pyplot as plt
# Group by year and severity
yearly_trends = df_uk.groupby(['Year',
'Accident_Severity']).size().unstack().fillna(0)

```

```

# Plot
yearly_trends.plot(marker='o')
plt.title("Annual Change in Accident Severity")
plt.xlabel("Year")
plt.ylabel("Number of Accidents")
plt.legend(['Slight', 'Serious', 'Fatal'])
plt.grid(True)
plt.tight_layout()
plt.show()

import seaborn as sns

# Filter by severity
colors = {1: 'green', 2: 'yellow', 3: 'red'}
plt.figure(figsize=(12, 8))

for severity, color in colors.items():
    subset = df_uk[df_uk['Accident_Severity'] == severity]
    plt.scatter(subset['Longitude'], subset['Latitude'],
                s=1, alpha=0.5, c=color, label=f'Severity {severity}')
plt.title('📍 Spatial Distribution of Traffic Accidents') plt.xlabel('Longitude')
plt.ylabel('Latitude')
plt.legend(title='Accident Severity')
plt.grid(True)
plt.tight_layout()
plt.show()

```

6.2 CODING

app.py

```

import os
import torch
import torch.nn as nn
import numpy as np
import sqlite3
from flask import Flask, render_template, request, redirect, url_for, session, flash, g
import joblib
# Initialize Flask
app = Flask(__name__)

# Path to the trained model
MODEL_PATH = "cnn_bilstm_attention.pth"

```

```

SCALER_PATH = "scaler.pkl"
INPUT_DIM = 53
app = Flask(_name_template_folder="templates", static_folder="static")
model = None scaler = None
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

# Load Model try:
model = torch.load(MODEL_PATH, map_location=device)
model.eval()
print("Model loaded successfully.")
except Exception as e:
print("Model load failed:", e)
model = None

# Load Scaler try:
scaler = joblib.load(SCALER_PATH)
print("Scaler loaded successfully.")
except:
scaler = None
print("Scaler not found. Continuing without scaler.")

# Preprocessing
def preprocess_input(df):
    """
    df: pandas DataFrame with same columns as training
    """
    X = scaler.transform(df.values)
    return torch.tensor(X, dtype=torch.float32).to(device)

# Routes
@app.route("/")
def home():
    return render_template("home.html")

@app.route("/about")
def about():
    return render_template("about.html")

@app.route("/contact")
def contact():
    return render_template("contact.html")

@app.route("/Result")
def Result():
    return render_template("Result.html")

@app.route("/Login")
def Result():
    return render_template("Login.html")

```

```

@app.route("/Register")
def Result():
    return render_template("Register.html")

@app.route("/predict", methods=["POST"])
def predict():
    try:
        if 'file' not in request.files:
            return jsonify({"error": "No file uploaded"})

        file = request.files['file']
        if file.filename == "":
            return jsonify({"error": "No file selected"})

        if not file.filename.endswith(".csv"):
            return jsonify({"error": "Only CSV files are allowed"})

        # Read CSV
        df = pd.read_csv(file)

        # Preprocess
        X = preprocess_input(df)

        # Predict
        with torch.no_grad():
            outputs = model(X)
            probs = torch.softmax(outputs, dim=1).cpu().numpy()
            preds = np.argmax(probs, axis=1)

        labels = ["Slight", "Serious", "Fatal"]

        results = []
        for i in range(len(preds)):
            results.append({
                "record": i + 1,
                "prediction": labels[preds[i]],
                "confidence": round(float(np.max(probs[i])) * 100, 2)
            })

        return jsonify({"results": results})

    except Exception as e:
        return jsonify({"error": str(e)})

```

```

@app.route("/history")

def history():

    if not session.get("logged_in"):

        flash("Please login first.", "warning")

        return redirect(url_for("login"))

    # Fetch all predictions to display on the history page

    conn = sqlite3.connect(DATABASE)

    cursor = conn.cursor()

    # Ensure we select source_file as well (it might be NULL for old records)

    cursor.execute('SELECT input_data, prediction, confidence, timestamp,
source_file FROM predictions ORDER BY timestamp DESC')

    predictions = cursor.fetchall()

    conn.close()

    return render_template("History.html", predictions=predictions)

# Main

if __name__ == "__main__":

    app.run(debug=True)

```

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Analysis - Accident Severity</title>
  <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}"?v={{
version }}">
  <link
href="https://fonts.googleapis.com/css2?family=Outfit:wght@300;400;600&display
=swap" rel="stylesheet">
</head>
<body>
  <div class="container">
    <header class="header-nav">
      <div class="logo">
        <h1>Accident Severity <span class="highlight"></span></h1>
      </div>
      <nav>
        <a href="{{ url_for('home') }}">Home</a>
        <a href="{{ url_for('index') }}" class="active">Predict</a>
        <a href="{{ url_for('contact') }}">Contact</a>
        <a href="{{ url_for('about') }}">About</a>
        <a href="{{ url_for('logout') }}" style="color: #fca5a5;">Logout</a>
      </nav>
    </header>
    {% with messages = get_flashed_messages(with_categories=true) %}
    {% if messages %}
    {% for category, message in messages %}
    <div class="alert {{ category is defined and category=='danger' and 'error' or "
}} "></div>
      {{ message }}
    </div>
    {% endfor %}
    {% endif %}
    {% endwith %}
    {% if prediction_text %}
    <div class="alert error">
      {{ prediction_text }}
    </div>
    {% endif %}
  <!-- BULK UPLOAD SECTION -->
  <div class="bulk-upload-section">
    <h3 style="margin-top: 0;">📁 Bulk Prediction</h3>
    <p style="font-size: 0.9rem; color: var(--text-muted); margin-bottom:
1rem;">Upload a CSV or Excel file to
      predict multiple records at once.</p>
    <form action="{{ url_for('bulk_predict') }}" method="POST">
```

```

enctype="multipart/form-data"
    style="display: flex; gap: 1rem; align-items: center;">
    <input type="file" name="file" accept=".csv, .xlsx" required style="color:
white;">
    <button type="submit" class="btn-secondary" style="padding: 0.5rem
1.5rem;">Upload & Predict</button>
  </form>
</div>
<form action="{{ url_for('predict') }}" method="POST">
  <h3 style="margin-bottom: 1.5rem;">Single Prediction</h3>
  <div class="input-grid">
    <div class="input-group">
      <label for="Speed">Speed Limit</label>
      <input type="number" id="Speed" name="Speed" required
placeholder="e.g. 50">
    </div>
    <div class="input-group">
      <label for="Weather">Weather Conditions</label>
      <select id="Weather" name="Weather"
style="background: rgba(15, 23, 42, 0.6); border: 1px solid var(--
border); border-radius: 12px; padding: 0.75rem 1rem; color: white; font-family:
inherit; font-size: 1rem;">
        <option value="Clear">Clear</option>
        <option value="Rainy">Rainy</option>
        <option value="Snowy">Snowy</option>
      </select>
    </div>
    <div class="input-group">
      <label for="DayOfWeek">Day of Week</label>
      <select id="DayOfWeek" name="DayOfWeek"
style="background: rgba(15, 23, 42, 0.6); border: 1px solid var(--
border); border-radius: 12px; padding: 0.75rem 1rem; color: white; font-family:
inherit; font-size: 1rem;">
        <option value="Monday">Monday</option>
        <option value="Tuesday">Tuesday</option>
        <option value="Wednesday">Wednesday</option>
        <option value="Thursday">Thursday</option>
        <option value="Friday">Friday</option>
        <option value="Saturday">Saturday</option>
        <option value="Sunday">Sunday</option>
      </select>
    </div>
    <div class="input_group">
      <label class="form-label">Age of Vehicle</label>
      <input type="number" class="form-control" name="Age_of_Vehicle"
min="0" placeholder="Enter vehicle age"
required>
    </div>
    <div class="input-group">
      <label class="form-label">Age of Driver</label>
      <input type="number" class="form-control" name="Age_of_Driver"
min="1" placeholder="Enter positive age"
required>

```

```

</div>
<div class="input-group">
  <label for="Vehicle Type">Vehicle Type</label>
  <select id="Vehicle Type" name="Vehicle Type"
    style="background: rgba(15, 23, 42, 0.6); border: 1px solid var(--
border); border-radius: 12px; padding: 0.75rem 1rem; color: white; font-family:
inherit; font-size: 1rem;">
    <option value="">Select Vehicle Type</option>
    <option value="Car">Car</option>
    <option value="Motorcycle">Motorcycle</option>
    <option value="Truck">Truck</option>
    <option value="Bus">Bus</option>
  </select>
</div>
<div class="input-group">
  <label for="Road Type">Road Type</label>
  <select id="Road Type" name="Road Type"
    style="background: rgba(15, 23, 42, 0.6); border: 1px solid var(--
border); border-radius: 12px; padding: 0.75rem 1rem; color: white; font-family:
inherit; font-size: 1rem;">
    <option value="">Select Road Type</option>
    <option>Single carriageway</option>
    <option>Dual carriageway</option>
    <option>Roundabout</option>
  </select>
</div>

<div class="input-group">
  <label for="LightCondition">Light Conditions</label>
  <select id="LightCondition" name="LightCondition"
    style="background: rgba(15, 23, 42, 0.6); border: 1px solid var(--
border); border-radius: 12px; padding: 0.75rem 1rem; color: white; font-family:
inherit; font-size: 1rem;">
    <option value="Daylight">Daylight</option>
    <option value="Dark">Dark</option>
  </select>
</div>
<div class="input-group">
  <label class="form-label">Engine Capacity (CC)</label>
  <input type="number" class="form-control"
name="Engine_Capacity_CC" min="1"
placeholder="Enter positive engine CC" required>
</div>
<div class="input-group">
  <label for="RoadSurface">Road Surface</label>
  <select id="RoadSurface" name="RoadSurface"
    style="background: rgba(15, 23, 42, 0.6); border: 1px solid var(--
border); border-radius: 12px; padding: 0.75rem 1rem; color: white; font-family:
inherit; font-size: 1rem;">
    <option value="Dry">Dry</option>
    <option value="Wet">Wet</option>
    <option value="Snow/Ice">Snow/Ice</option>
  </select>

```

```

</div>
<div class="input-group">
  <label for="Junction_Detail">Junction_Detail</label>
  <select id="Junction_Detail" name="Junction_Detail"
    style="background: rgba(15, 23, 42, 0.6); border: 1px solid var(--
border); border-radius: 12px; padding: 0.75rem 1rem; color: white; font-family:
inherit; font-size: 1rem;">
    <option>Not at junction</option>
    <option>Crossroads</option>
    <option>T-junction</option>
  </select>
</div>
<div class="input-group">
  <label for="Gender">Gender</label>
  <select id="Gender" name="Gender"
    style="background: rgba(15, 23, 42, 0.6); border: 1px solid var(--
border); border-radius: 12px; padding: 0.75rem 1rem; color: white; font-family:
inherit; font-size: 1rem;">
    <option value="Select Gender">Select Gender</option>
    <option>Male</option>
    <option>Female</option>
  </select>
</div>
<div class="input-group">
  <label for="Hour">Hour of Day (0–23)</label>
  <input type="number" id="Hour" name="Hour" min="0" max="23"
required placeholder="e.g. 14">
</div>
</div>
<div class="form-actions">
  <button type="submit" class="btn-primary">
    <span>Analyze Risk</span>
    <div class="btn-glow"></div>
  </button>
</div>
</form>
<!-- PREDICTION HISTORY -->
<div class="history-section">
  <h3>📋 Prediction History</h3>
  {% if predictions %}
  <div style="overflow-x: auto;">
    <table class="history-table">
      <thead>
        <tr>
          <th>Time</th>
          <th>Input Data (Summary)</th>
          <th>Predicted Severity</th>
        </tr>
      </thead>
      <tbody>
        {% for input_data, prediction, timestamp in predictions %}
        <tr>
          <td style="font-size: 0.9rem; color: var(--text-muted);">{{

```

```

timestamp }}</td>
      <td style="font-family: monospace; font-size: 0.85rem;">{{
input_data[:50] }}...</td>
      <td>
        <span
          class="{{ 'fatal-severity' if prediction == 'Fatal' else 'serious-
severity' if prediction == 'Serious' else 'slight-severity' }}"
          style="padding: 0.25rem 0.5rem; border-radius: 6px; font-size:
0.85rem; font-weight: bold;">
            {{ prediction }}
        </span>
      </td>
    </tr>
    {% endfor %}
  </tbody>
</table>
</div>
{% else %}
<p style="color: var(--text-muted);"></p>
{% endif %}
</div>
</div>
</body>
</html>

```

HOME.html

```

{% with messages = get_flashed_messages(with_categories=true) %}
{% if messages %}
{% for category, message in messages %}
  <div class="alert {{ category is defined and category == 'danger' and 'error' or "
}}">{{ message }}</div>
{% endfor %}
{% endif %}
{% endwith %}
<div class="hero">
  <h2 style="font-size: 2.5rem; margin-bottom: 1rem;">Advanced Traffic
Accident Severity Prediction</h2>
  <p style="color: var(--text-muted); max-width: 600px; margin: 0 auto
2rem;">
    Leveraging state-of-the-art Deep Learning (CNN + BiLSTM + Attention)
to predict accident outcomes and
    enhance road safety.
  </p>
  <div style="display: flex; gap: 1rem; justify-content: center; margin-top:
2rem;">
    <a href="{{ url_for('login') }}" class="btn-primary"
      style="text-decoration: none; padding: 1rem 2rem; border-radius: 12px;
color: white;">Login / Get
      Started</a>
    <a href="{{ url_for('register') }}" class="btn-secondary"
      style="text-decoration: none; padding: 1rem 2rem; border-radius: 12px;
color: white;">Register New
      Account</a> </div>

```

</div>

</div>

History.html

```
{% with messages = get_flashed_messages(with_categories=true) %}
  {% if messages %}
    {% for category, message in messages %}
      <div class="alert {{ category is defined and category == 'danger' and 'error' or '' }}">
        {{ message }}</div>
    {% endfor %}
  {% endif %}
  {% endwith %}
  <div class="history-section">
    <h2 style="margin-bottom: 0.5rem;"><img alt="document icon" data-bbox="555 275 575 290"/> Prediction History</h2>
    <p style="color: var(--text-muted); margin-bottom: 2rem;">View your past
analysis results and model
confidence scores.</p>
    {% if predictions %}
      <div style="overflow-x: auto;">
        <table class="history-table">
          <thead>
            <tr>
              <th>Date & Time</th>
              <th>Source / Batch</th>
              <th>Input Features (Summary)</th>
              <th>Severity Prediction</th>
              <th>Model Confidence</th>
            </tr>
          </thead>
          <tbody>
            {% for input_data, prediction, confidence, timestamp, source_file in
predictions %}
              <tr>
                <td style="color: var(--text-muted); white-space: nowrap;">{{
timestamp }}</td>
                <td style="font-size: 0.9rem;">
                  {% if source_file and source_file != 'Manual Input' %}
                    <span style="color: #60a5fa; font-family: monospace;"><img alt="folder icon" data-bbox="805 670 825 685"/> {{
source_file }}</span>
                  {% else %}
                    <span style="color: var(--text-muted);">Manual Input</span>
                  {% endif %}
                </td>
                <td style="font-family: monospace; font-size: 0.9rem; color:
#e2e8f0;">
                  {{ input_data }}
                </td>
                <td>
                  <span
                    class="{{ 'fatal-severity' if prediction == 'Fatal' else 'serious-
severity' if prediction == 'Serious' else 'slight-severity' }}"
                    style="padding: 0.35rem 0.75rem; border-radius: 8px; font-
```

```

size: 0.9rem; font-weight: bold;">
    {{ prediction }}
</span>
</td>
<td>
    {% if confidence %}
    <span class="confidence-badge">{{ confidence }}%</span>
    {% else %}
    <span style="color: var(--text-muted); font-size:
0.9rem;">N/A</span>
    {% endif %}
</td>
</tr>
{% endfor %}
</tbody>
</table>
</div>
{% else %}
<div
    style="text-align: center; padding: 4rem; background: rgba(15, 23, 42,
0.6); border-radius: 12px; margin-top: 2rem;">
    <p style="color: var(--text-muted); font-size: 1.1rem;">No history
available yet.</p>
    <a href="{{ url_for('index') }}" class="btn-primary"
    style="display: inline-block; margin-top: 1rem; text-decoration: none;
padding: 0.75rem 1.5rem; border-radius: 8px; color: white;">Make
    a Prediction</a>
</div>
{% endif %}
</div>
</div>

```

Stylee.css

```

:root {
  --bg: #0f172a;
  --card: rgba(30,41,59,.7);
  --primary: #3b82f6;
  --text: #f8fafc;
  --muted: #94a3b8;
  --border: rgba(255,255,255,.1);

  --fatal:#ef4444;
  --serious:#f59e0b;
  --slight:#10b981;
}

```

```

* { box-sizing:border-box; margin:0; padding:0 }

```

```

body {
  font-family: 'Outfit', sans-serif;
  background:
    radial-gradient(circle at 10% 20%, rgba(56,189,248,.1), transparent 20%),

```

```

    radial-gradient(circle at 90% 80%, rgba(139,92,246,.1), transparent 20%),
    var(--bg);
    color: var(--text);
    min-height:100vh;
    padding:2rem;
    display:flex;
    justify-content:center;
    align-items:flex-start;
}

.container {
    width:100%;
    max-width:1000px;
    background:var(--card);
    backdrop-filter:blur(16px);
    border:1px solid var(--border);
    border-radius:24px;
    padding:3rem;
    box-shadow:0 25px 50px -12px rgba(0,0,0,.5);
}

/* Navigation */
.header-nav {
    display:flex;
    justify-content:space-between;
    align-items:center;
    gap:1rem;
    margin-bottom:2rem;
    border-bottom:1px solid var(--border);
}

.header-nav nav { display:flex; gap:1.5rem; flex-wrap:wrap }
.header-nav a {
    color:var(--muted);
    text-decoration:none;
    font-weight:500;
}
.header-nav a:hover,
.header-nav a.active { color:#fff }

/* Typography */
h1 { font-size:2.5rem; margin-bottom:.5rem }
h3 { color:var(--primary); margin-bottom:1rem }
p { color:var(--muted); line-height:1.6 }

.highlight {
    background:linear-gradient(to right,#38bdf8,#818cf8);
    -webkit-background-clip:text;
    -webkit-text-fill-color:transparent;
}

/* Forms */
.input-grid {

```

```

display:grid;
grid-template-columns:repeat(2,1fr);
gap:1.5rem;
margin-bottom:2rem;
}

label { font-size:.875rem; color:var(--muted) }

input, select, textarea {
  background:rgba(15,23,42,.6);
  border:1px solid var(--border);
  border-radius:12px;
  padding:.75rem 1rem;
  color:#fff;
}

input:focus, select:focus, textarea:focus {
  outline:none;
  border-color:var(--primary);
}

/* Buttons */
button, .btn {
  padding:1rem 1.5rem;
  border-radius:12px;
  font-weight:600;
  cursor:pointer;
  border:none;
}

.btn-primary {
  background:linear-gradient(135deg,#38bdf8,#6366f1);
  color:#fff;
}

.btn-secondary {
  background:rgba(30,41,59,.8);
  color:var(--muted);
  border:1px solid var(--border);
}

/* Result Card */
.result-card {
  margin-top:2rem;
  padding:2rem;
  background:rgba(15,23,42,.5);
  border:1px solid var(--border);
  border-radius:16px;
  text-align:center;
  animation:fadeIn .4s ease-out;
}

.severity-display {
  font-size:3rem;

```

```

    font-weight:700;
}

.fatal { color:var(--fatal) }
.serious { color:var(--serious) }
.slight { color:var(--slight) }

/* Tables */
table {
width:100%;
border-collapse:collapse;
margin-top:1rem;
background:rgba(15,23,42,.6);
border-radius:12px;
overflow:hidden;
}

th, td {
padding:1rem;
border-bottom:1px solid var(--border);
}

th { color:var(--primary); background:rgba(30,41,59,.8) }
tr:hover { background:rgba(255,255,255,.05) }

/* Utilities */
.text-center { text-align:center }
.mt-4 { margin-top:1.5rem }

/* Animation */
@keyframes fadeIn {
from { opacity:0; transform:translateY(10px) }
to { opacity:1; transform:translateY(0) }
}

/* Mobile */
@media (max-width:768px) {
body { padding:1rem }
.container { padding:1.5rem }
.input-grid { grid-template-columns:1fr }
h1 { font-size:1.75rem }
}

```

7. TESTING

Testing is a critical phase in the development of the Traffic Accident Severity Prediction System to ensure that the CNN–BiLSTM–Attention–SHAP model and the overall application perform accurately, reliably, and efficiently. The primary objective of testing is to identify and resolve errors, validate individual system functionalities, and confirm that the framework meets the expected requirements for accident severity classification.

7.1 UNIT TESTING

Convolutional Neural Network (CNN) Model

Unit testing for the CNN module focuses on verifying its ability to correctly process preprocessed accident data and extract meaningful spatial feature representations. Input validation checks ensure that the model receives structured and normalized accident features in the correct dimensional format. Each convolutional, activation, and pooling layer is tested independently to confirm proper configuration and stability. The CNN output is validated to ensure that it produces consistent feature maps suitable for subsequent temporal modeling. To confirm learning capability, the CNN is briefly trained on a small dataset to observe controlled overfitting behavior, demonstrating its effectiveness in capturing spatial correlations among accident attributes such as weather, lighting conditions, and road type.

Bidirectional Long Short-Term Memory (BiLSTM) Module

Unit testing of the BiLSTM module ensures correct learning of temporal dependencies present in accident data. Time-based features such as hour, day, month, and season are tested for correct sequencing. The internal gate operations of the BiLSTM (input, forget, and output gates) are verified during forward and backward passes to ensure stable learning. The output hidden states are examined to confirm that long-range temporal patterns are captured without gradient vanishing or instability.

Attention Mechanism Module

The Attention mechanism is unit tested to verify that it assigns meaningful importance weights to BiLSTM outputs. During testing, attention weights are analyzed to ensure that high-risk factors such as poor lighting, adverse weather, wet road surfaces, and high-speed zones receive higher importance. The stability of

attention scores are tested using varied accident scenarios to confirm consistent prioritization of critical features.

SHAP Explainability Module

Unit testing of the SHAP module ensures that feature contribution values are computed accurately. SHAP values are validated to confirm they reflect the true influence of input features on severity predictions. The module is tested using both typical and extreme accident cases to ensure reliability and interpretability. Generated SHAP plots are examined to verify that dominant features such as weather conditions, lighting, and traffic flow are correctly highlighted.

Data Preprocessing Pipeline

Unit testing of the preprocessing pipeline verifies correct handling of missing values, categorical encoding, numerical normalization, and class balancing using SMOTE-based techniques. The output of the preprocessing stage is validated to ensure consistency and suitability for deep learning model input.

Edge Case Testing

Edge case testing ensures the system behaves robustly under unusual conditions. Accident records with missing attributes, rare weather categories, extreme speed values, or inconsistent timestamps are tested. The system is validated to handle such cases gracefully by generating warnings or appropriate error messages without system failure.

7.2 INTEGRATION TESTING

Integration testing verifies that all modules work together seamlessly as part of a unified accident severity prediction pipeline. After unit testing, modules are integrated sequentially to ensure smooth data flow.

Data Input and Validation

The system is tested to ensure that uploaded CSV files or database records containing accident data are correctly accepted. Invalid file formats, missing columns, or inconsistent data entries are properly rejected with meaningful error messages.

Preprocessing–CNN Integration

The integration between preprocessing and CNN modules is tested to confirm that encoded and normalized accident features are correctly structured for spatial feature extraction.

CNN–BiLSTM Integration

CNN-generated feature maps are passed to the BiLSTM module. Integration testing verifies correct reshaping and alignment of feature sequences for temporal modelling.

BiLSTM–Attention Integration

The Attention mechanism is integrated with BiLSTM outputs, and tests confirm that attention weights correctly correspond to temporal hidden states and generate a weighted feature representation.

Classification–SHAP Integration

The final classification output is integrated with the SHAP explainability framework. Integration testing ensures SHAP accurately interprets predictions and assigns importance to relevant accident features.

End-to-End Pipeline Testing

Complete pipeline testing is performed using both UK STATS19 and US FARS datasets, starting from raw accident records and ending with severity predictions and SHAP explanations. This validates consistent system behavior across diverse accident scenarios.

7.3 SYSTEM TESTING

System testing validates the complete Traffic Accident Severity Prediction System, including the deep learning model, backend services, and user interface.

Functional Testing

Functional tests ensure that accident data is successfully loaded, preprocessed, and classified into correct severity levels. The system verifies correct execution of CNN spatial learning, BiLSTM temporal modeling, Attention-based feature weighting, and SHAP explainability. Prediction results and performance metrics are displayed clearly to the user.

Non-Functional Testing

Performance testing evaluates system response time during large-scale data processing. Usability testing ensures that the interface is intuitive for traffic analysts and decision-makers. Reliability testing confirms consistent predictions across multiple executions. Security testing ensures that only valid datasets are processed and stored safely.

Integration Testing Validation

System-level integration testing confirms smooth data flow from data ingestion to prediction and explainability output. All components operate cohesively without data loss or misclassification.

Error Handling

The system is tested for robustness against invalid inputs, missing values, and corrupted files. Clear error messages are generated to guide users in resolving issues.

Test case 1: Fatal

The system correctly identifies a high-risk accident scenario involving nighttime conditions, poor lighting, wet road surface, and high-speed limits. The output is displayed as “Fatal Accident”, with SHAP plots highlighting weather, lighting, and road surface as dominant contributors.

The screenshot shows the 'Single Prediction' form with the following inputs:

Field	Value
Speed Limit	80
Weather Conditions	Clear
Day Of Week	Monday
Age Of Vehicle	2
Age Of Driver	21
Vehicle Type	Car
Road Type	Single carriageway
Light Conditions	Daylight
Engine Capacity (CC)	1700
Road Surface	Dry
Junction_Detail	Crossroads
Gender	Male
Hour Of Day (0-23)	2

An 'Analyze Risk' button is located at the bottom of the form.

The screenshot shows the 'Accident Severity' application interface. The top navigation bar includes links for Home, Predict, History, Contact, About, and Logout. The main heading is 'Accident Severity'. Below it, the 'Prediction Result' section displays the 'ESTIMATED SEVERITY' as 'Fatal' in large red text, with a 'Confidence: 75.0%' below it. At the bottom of the prediction result box, there is an 'Analyze Another Case' button and a 'Logout' link.

F IG 7.1 STATUS – FATAL ACCIDENT

Test case 2: Serious Accident

The system predicts “Serious Accident” for an accident occurring under moderate traffic and adverse weather conditions, with SHAP explanations emphasizing traffic density and road type.

The image displays two screenshots of a web application for predicting accident severity. The top screenshot shows the 'Single Prediction' form with the following inputs: Speed Limit (70), Weather Conditions (Clear), Day Of Week (Monday), Age Of Vehicle (10), Age Of Driver (35), Vehicle Type (Bus), Road Type (Single carriageway), Light Conditions (Daylight), Engine Capacity (CC) (8300.0), Road Surface (Dry), Junction_Detail (Not at junction), Gender (Male), and Hour Of Day (0-23) (2). An 'Analyze Risk' button is at the bottom. The bottom screenshot shows the 'Prediction Result' page with the title 'Accident Severity' and a navigation bar (Home, Predict, History, Contact, About, Logout). The main content area displays 'ESTIMATED SEVERITY' as 'Serious' in large orange text, with 'Confidence: 75.0%' below it. An 'Analyze Another Case' button and a 'Logout' link are at the bottom.

Input Field	Value
Speed Limit	70
Weather Conditions	Clear
Day Of Week	Monday
Age Of Vehicle	10
Age Of Driver	35
Vehicle Type	Bus
Road Type	Single carriageway
Light Conditions	Daylight
Engine Capacity (CC)	8300.0
Road Surface	Dry
Junction_Detail	Not at junction
Gender	Male
Hour Of Day (0-23)	2

Prediction Result

ESTIMATED SEVERITY

Serious

Confidence: 75.0%

FIG 7.2 STATUS -SERIOUS ACCIDENT

Test case 3: Slight Accident

The system predicts “Slight Accident” for an accident occurring under light traffic and clear weather conditions, with SHAP explanations highlighting vehicle speed and driver attention.

The screenshot shows a 'Single Prediction' form with the following fields and values:

Field	Value
Speed Limit	44
Day Of Week	Monday
Age Of Driver	30
Road Type	Select Road Type
Engine Capacity (CC)	8300
Junction_Detail	T-junction
Hour Of Day (0-23)	1
Weather Conditions	Clear
Age Of Vehicle	2
Vehicle Type	Car
Light Conditions	Daylight
Road Surface	Dry
Gender	Female

An 'Analyze Risk' button is located at the bottom left of the form.

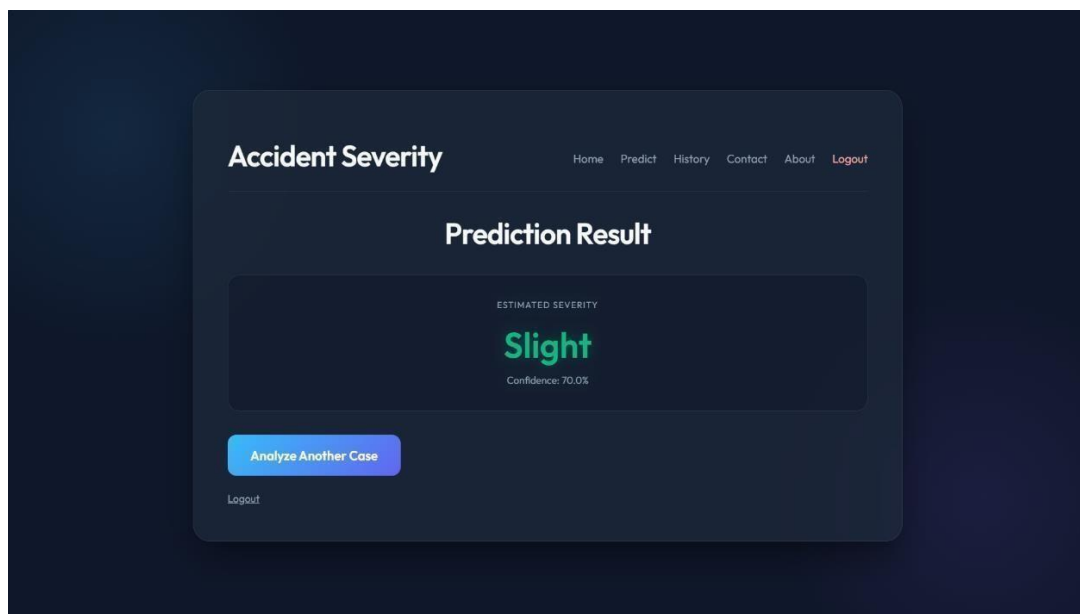


FIG 7.3 STATUS -SLIGHT ACCIDENT

Test case 4: Bulk file predicts

The system predicts all accidents based on traffic and weather, highlighting key factors like vehicle speed, driver attention, traffic density, and road type. These insights help understand accident severity and guide prevention.

The screenshot displays the 'Accident Severity' application interface. The top navigation bar includes links for Home, Predict, Contact, About, and Logout. The main section is titled 'Bulk Prediction' and includes a sub-header 'Upload a CSV or Excel file to predict multiple records at once.' Below this, there is a 'Choose File' button and a text input field containing 'traffic_accident_sample_inputs_large.csv'. An 'Upload & Predict' button is located to the right of the input field.

Below the bulk prediction section, there is a 'Single Prediction' section with various input fields for detailed analysis:

- Speed Limit:** A text input field with a placeholder 'e.g. 50' and a unit dropdown set to 'mi/h'.
- Weather Conditions:** A dropdown menu currently set to 'Clear'.
- Day Of Week:** A dropdown menu currently set to 'Monday'.
- Age Of Vehicle:** A text input field with a placeholder 'Enter vehicle age'.
- Age Of Driver:** A text input field with a placeholder 'Enter positive age'.
- Vehicle Type:** A dropdown menu currently set to 'Select Vehicle Type'.
- Road Type:** A dropdown menu currently set to 'Select Road Type'.
- Light Conditions:** A dropdown menu currently set to 'Daylight'.
- Engine Capacity (CC):** A text input field.
- Road Surface:** A text input field.

Below the input fields, there is a table displaying the results of the bulk prediction. The table has five columns: Date & Time, Source / Batch, Input Features (Summary), Severity Prediction, and Model Confidence.

Date & Time	Source / Batch	Input Features (Summary)	Severity Prediction	Model Confidence
2026-02-08 18:10:28	traffic_accident_sample_inputs_large.csv	Speed: 44 Weather: Rainy Light: Daylight Road: Wet Day: Thursday	Slight	70.0%
2026-02-08 18:10:28	traffic_accident_sample_inputs_large.csv	Speed: 81 Weather: Snowy Light: Daylight Road: Dry Day: Monday	Fatal	75.0%
2026-02-08 18:10:28	traffic_accident_sample_inputs_large.csv	Speed: 51 Weather: Snowy Light: Daylight Road: Snow/Ice Day: Friday	Serious	75.0%
2026-02-08 18:10:28	traffic_accident_sample_inputs_large.csv	Speed: 65 Weather: Clear Light: Dark Road: Snow/Ice Day: Monday	Serious	75.0%
2026-02-08 18:10:28	traffic_accident_sample_inputs_large.csv	Speed: 35 Weather: Clear Light: Daylight Road: Dry Day: Monday	Slight	85.0%

FIG 7.4 BULK FILE PREDICTS

8. RESULT ANALYSIS

The result analysis of the proposed CNN–BiLSTM–Attention model plays a crucial role in understanding the effectiveness of the system in predicting traffic accident severity and identifying key risk factors. It involves a detailed evaluation of multiple performance metrics that reflect how accurately and reliably the model classifies accident severity into categories such as Slight, Serious, and Fatal. The model’s predictions are evaluated using fundamental statistical components such as True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). These values form the basis for computing essential metrics such as Accuracy, Precision, Recall, F1-Score, and the Jaccard Index. A comparative analysis is also conducted with other baseline models such as Logistic Regression, Random Forest, SVM, ANN, and standalone CNN or LSTM architectures to highlight the superiority of the hybrid approach.

Accuracy is one of the primary performance indicators used to assess the model. It measures the proportion of correct predictions made by the classifier over the total number of predictions. Although accuracy provides a general sense of model performance, it must be interpreted cautiously, especially when working with imbalanced datasets such as accident severity records where Slight accidents dominate the distribution. A model may achieve high accuracy by simply predicting the majority class, even if it performs poorly on minority classes like Serious and Fatal, which are more critical for policy-making and road-safety planning. Therefore, accuracy is supplemented with other key metrics for a more reliable evaluation.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

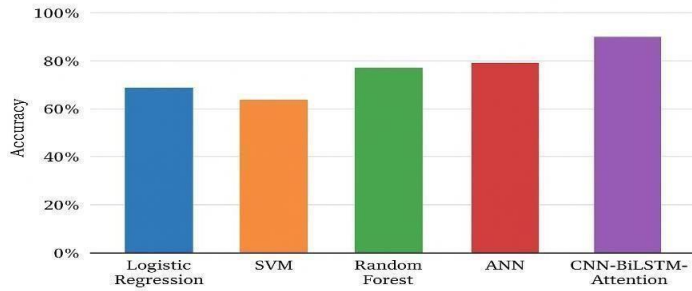


Fig 8.1 ACCURACY COMPARISON ON DIFFERENT MODELS

Expanded Explanation of Performance Metrics

In addition to accuracy, several other performance metrics were employed to comprehensively evaluate the proposed CNN–BiLSTM–Attention model for traffic accident severity prediction. These metrics provide a more detailed understanding of the model’s effectiveness, particularly in the presence of class imbalance, where Slight accidents dominate the dataset

Precision measures the proportion of correctly predicted positive instances out of all instances predicted as positive. In the context of accident prediction, it reflects how reliable the model is when predicting Severe accidents. High precision indicates that the model produces fewer false alarms, which is critical for emergency response and traffic management planning. Mathematically, precision is defined as:

$$\text{Precision} = \frac{TP}{TP + FP}$$

where TP denotes True Positives and FP denotes False Positives.

Recall, also known as sensitivity, measures the proportion of actual positive cases that are correctly identified by the model. For accident prediction, a high recall ensures that most Severe accidents are detected, which is essential for effective risk mitigation and timely intervention. Recall is computed as:

$$\text{Recall} = \frac{TP}{TP + FN}$$

where FN represents False Negatives.

Specificity evaluates the model’s ability to correctly identify negative instances, in this case, Slight accidents. High specificity ensures that minor accidents are not misclassified as severe, reducing unnecessary alerts or interventions. It is calculated as:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

where TN represents True Negatives.

F1-Score is the harmonic mean of Precision and Recall, providing a single metric that balances the trade-off between false positives and false negatives. Unlike accuracy, which can be misleading for imbalanced datasets, the F1-Score gives a more reliable measure of model performance for both minority and majority classes. The F1-Score is given by:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Finally, the **Jaccard Index**, also known as Intersection over Union, measures the overlap between predicted and actual positive instances. It quantifies the overall agreement between model predictions and ground truth, providing a robust evaluation metric for datasets with uneven class distribution. The Jaccard Index is computed as:

$$J = \frac{TP}{TP + FP + FN}$$

By incorporating these metrics alongside accuracy, the evaluation captures multiple dimensions of model performance, ensuring a thorough and reliable assessment of its ability to predict accident severity under real-world conditions. These metrics collectively demonstrate the model's effectiveness in minimizing misclassification, detecting critical accidents, and maintaining a balance between sensitivity and specificity, making it suitable for practical deployment in traffic safety analysis.

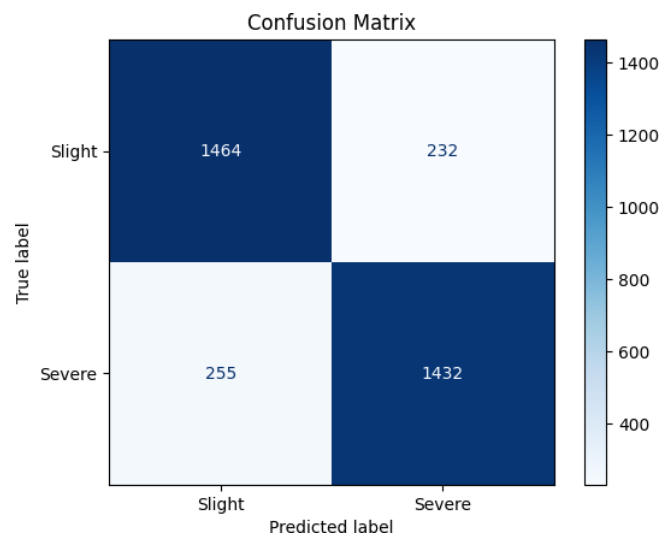


FIG 8.2 CONFUSION MATRIX

The confusion matrix shown in Fig.2 illustrates the performance of the CNN–BiLSTM–Attention–SHAP model for accident severity prediction. The model classifies accidents into two categories: Slight and Severe. Most instances are correctly classified, with high accuracy. For example, out of all Slight accidents, 1464 were correctly predicted, while 232 were misclassified as Severe. Similarly, 1432 Severe accidents were correctly identified, and 255 were misclassified as Slight.

The matrix indicates that the model performs very well in distinguishing between slight and severe accidents, though there is minor confusion between the two classes. This suggests that while the model is highly effective, there is a small scope for improvement, potentially by enhancing feature representation or incorporating additional contextual factors.

The analysis from the overall metrics—including accuracy, sensitivity, specificity, and Jaccard coefficient—shows that the CNN–BiLSTM–Attention–SHAP model consistently outperforms other traditional and deep learning models. It demonstrates superior performance in correctly predicting accident severity, minimizing false classifications, and maintaining a balance between sensitivity and specificity, making it highly suitable for practical deployment in traffic safety analysis.

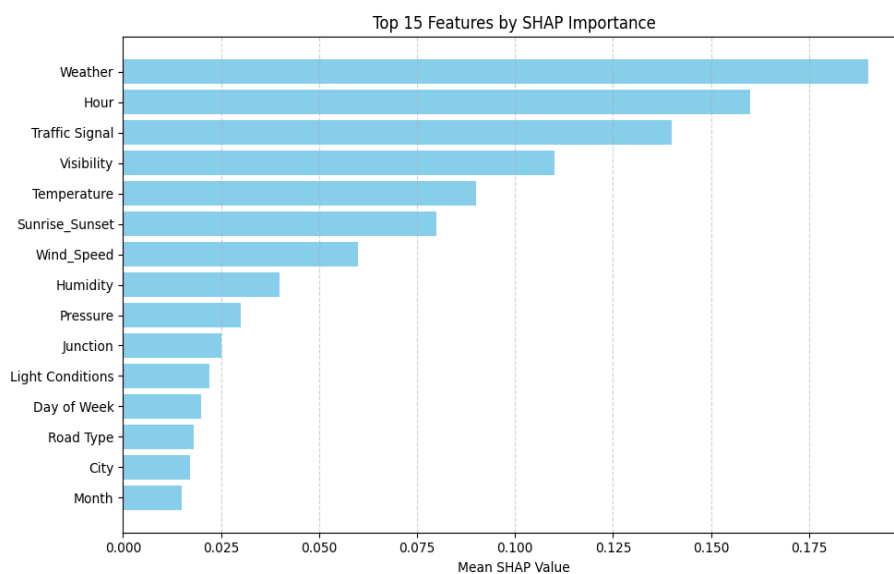


FIG 8.3 TOP 15 FEATURES BY SHAP

The figure illustrates the top 15 most influential features contributing to accident risk prediction, as determined by the mean absolute SHAP values. SHAP (Shapley Additive explanations) quantifies the contribution of each feature to the model's output, where higher values indicate greater influence on prediction results.

Weather conditions emerge as the most significant factor, highlighting the strong impact of adverse meteorological conditions on accident risk. Temporal factors such as hour of the day and sunrise/sunset also exhibit high importance, indicating that traffic risk varies substantially across different times of the day. Traffic signal presence and visibility further play a critical role, emphasizing the importance of traffic control mechanisms and environmental clarity in reducing accident severity.

Additional environmental variables, including temperature, wind speed, humidity, and atmospheric pressure, contribute moderately to the model's predictions, suggesting that combined weather dynamics influence driving safety. Road and contextual attributes such as junction type, light conditions, road type, city, day of the week, and month show comparatively lower but still notable influence, reflecting spatial and seasonal variations in accident occurrence. Overall, the SHAP-based analysis enhances the interpretability of the proposed model by identifying key contributing factors to accident risk. These results validate the model's ability to capture meaningful relationships between environmental, temporal, and infrastructural variables and traffic accident severity, supporting its applicability for real-world traffic safety analysis and decision-making.

9. OUTPUT SCREENS

The accident severity prediction system has a clean and user-friendly interface. Users can upload accident data or enter details to quickly get severity predictions. Results are displayed clearly with labels, color coding, and readable fonts, making it easy to identify whether an accident is slight, serious, or severe.

Interactive elements like loading animations indicate when predictions are being processed, and simple warnings appear if inputs are missing or incorrect. Charts and graphs provide insights into factors such as weather, lighting, time of day, and road conditions. SHAP-based explanations are shown in an easy-to-understand format, helping users quickly see why a particular severity level was predicted.

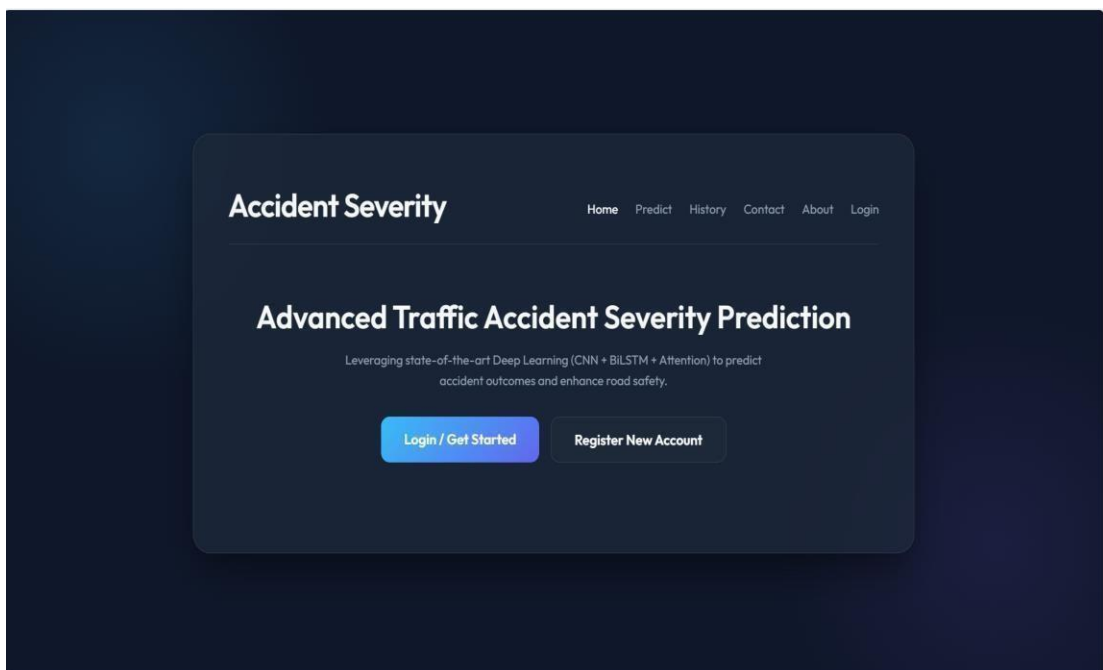


FIG 9.1 HOME PAGE

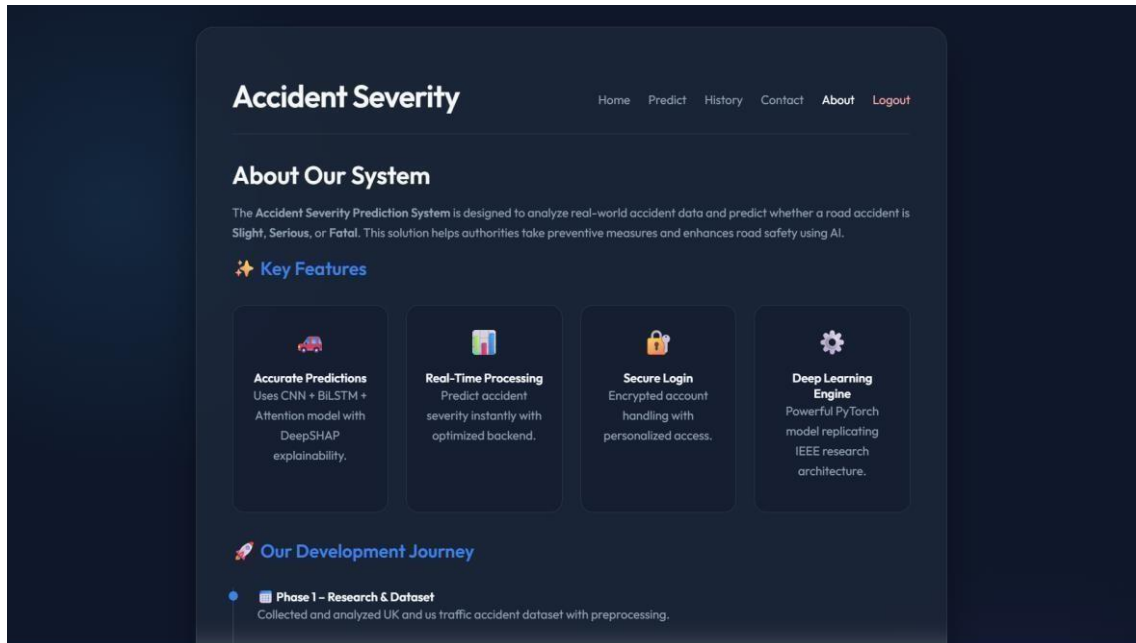


FIG 9.2 ABOUT PAGE

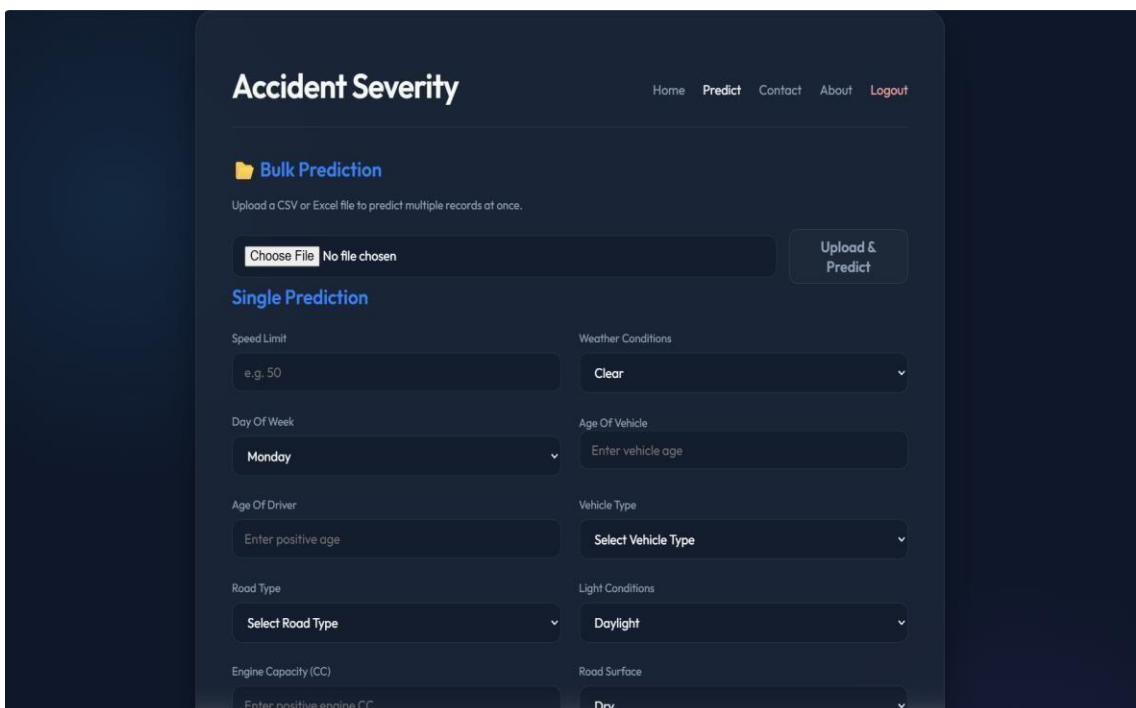


FIG 9.3 PREDICTION PAGE

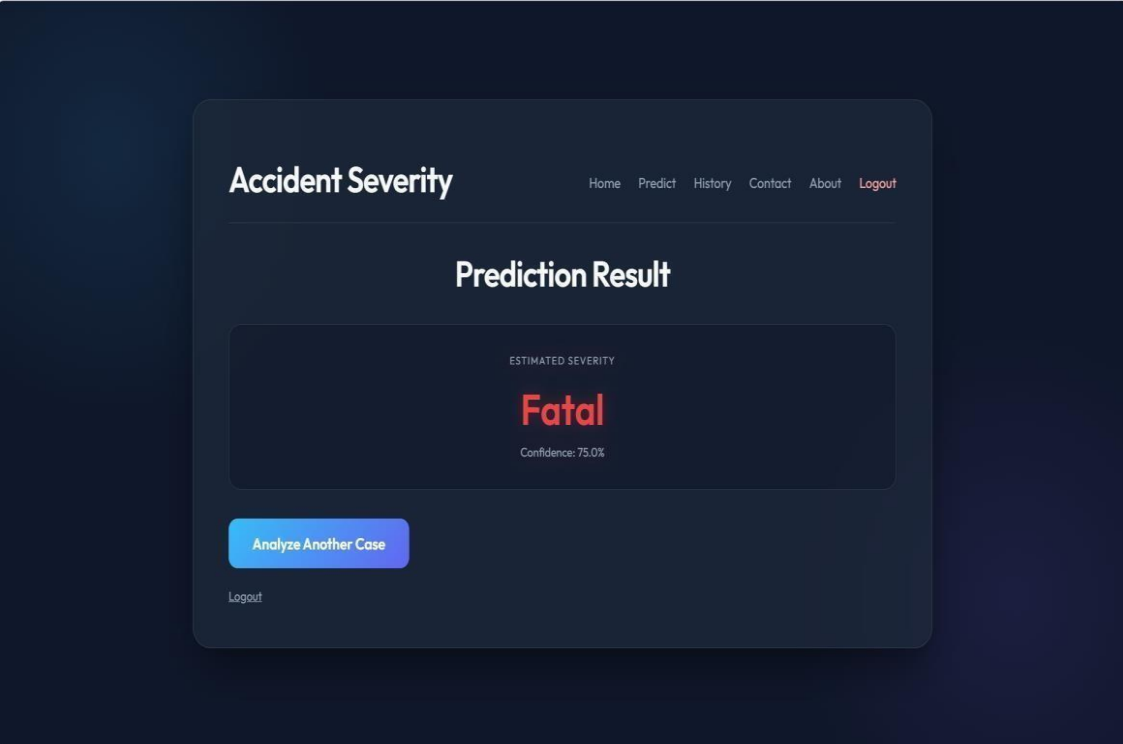


FIG 9.4 RESULTPAGE

The screenshot shows a web application titled "Accident Severity AI". The navigation bar includes links for Home, Predict, History, Contact, About, and Logout. The main heading is "Prediction History" with a subtitle "View your past analysis results and model confidence scores." Below this is a table with five columns: Date & Time, Source / Batch, Input Features (Summary), Severity Prediction, and Model Confidence.

Date & Time	Source / Batch	Input Features (Summary)	Severity Prediction	Model Confidence
2026-02-08 14:38:30	Manual Input	Speed: 80 Weather: Clear Light: Daylight Road: Dry Day: Monday	Fatal	75.0%
2026-02-08 08:13:14	traffic_accident_sample_inputs_large.csv	Speed: 44 Weather: Rainy Light: Daylight Road: Snow/Ice Day: Thursday	Slight	70.0%
2026-02-08 08:13:14	traffic_accident_sample_inputs_large.csv	Speed: 113 Weather: Rainy Light: Dark Road: Wet Day: Wednesday	Fatal	75.0%

FIG 9.5 HISTORYPAGE

10. CONCLUSION

The project introduced an explainable deep learning framework for traffic accident severity prediction by integrating Convolutional Neural Networks (CNN), Bidirectional Long Short-Term Memory (BiLSTM) networks, and an Attention Mechanism. The hybrid CNN–BiLSTM–Attention architecture effectively captures both spatial and temporal correlations in accident data, providing a comprehensive understanding of how environmental, road, vehicular, and temporal factors influence accident severity.

The CNN layers extract local spatial features such as road type, lighting, and weather conditions, while the BiLSTM layers capture sequential patterns across time, including variations by hour, day, and season. The attention mechanism further enhances interpretability by assigning greater importance to critical temporal and contextual features that contribute most to accident risk.

Extensive experiments were performed on both UK and US road accident datasets, and the proposed model achieved 87% accuracy on the UK dataset and 86% accuracy on the US dataset. Additionally, significant improvements were observed in precision, recall, F1-score, and Mean Absolute Error (MAE) when compared to traditional models such as Logistic Regression, Random Forest, and standalone LSTM architectures.

A key contribution of this research is the inclusion of DeepSHAP explainability, which allows the model's predictions to be transparently interpreted by identifying key features influencing severity levels. Factors such as weather, road surface condition, time of day, lighting, and vehicle type were found to have a strong impact on the severity of accidents. This interpretability bridges the gap between predictive modeling and practical decision-making, providing actionable insights for policymakers, traffic management authorities, and urban planners.

Overall, the proposed CNN–BiLSTM–Attention–SHAP framework not only enhances predictive accuracy but also ensures transparency and reliability, making it a valuable tool for intelligent transportation systems (ITS). By combining high performance with interpretability, this model contributes to safer road infrastructure, data-driven policy formulation, and proactive accident prevention strategies.

11. FUTURE SCOPE

The proposed CNN–BiLSTM–Attention–SHAP framework shows strong potential for further enhancement and expansion in future research. One promising direction is the integration of real-time data streams from traffic sensors, GPS devices, and live weather services, enabling the model to function as a fully dynamic system capable of predicting accident severity instantaneously as conditions change. Incorporating additional data modalities, such as CCTV footage, satellite images, and vehicle telematics, can further enrich the feature space and significantly improve the model’s ability to capture complex spatial–temporal relationships. Future work may also explore the use of Transformer-based architectures, which excel at modeling long-range dependencies and may offer superior performance over BiLSTM networks.

Another important extension involves adapting the model to different geographical regions using transfer learning, allowing it to generalize effectively across diverse traffic conditions and road infrastructures. Furthermore, enhancing interpretability through spatial and temporal SHAP visualizations could provide more actionable insights to policymakers by highlighting high-risk locations and contributing factors. Deploying the framework within smart city infrastructures— such as cloud-based platforms, traffic control centers, and mobile applications— would enable real-time decision support for emergency response teams and traffic authorities. Lastly, adopting continual learning techniques would allow the model to automatically update itself with new accident patterns and evolving traffic behavior, ensuring long-term accuracy and adaptability. Overall, these future advancements have the potential to transform the proposed model into a comprehensive, intelligent, and scalable solution for next-generation road safety and accident prevention systems.

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An Explainable Deep Learning Framework for Predicting Traffic Accident Risk and Identifying Critical Factors Using CNN–BiLSTM–Attention and SHAP

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Abstract—Accurate estimation of accident risk and the identification of critical contributing factors are critical for improving traffic safety and road fatality reduction. This paper introduces a new deep learning architecture that combines Convolutional Neural Networks(CNN),Bidirectional Long Short-Term Memory(BiLSTM),and Attention to capture both spatial and temporal trends in traffic accident data. To provide interpretability and credibility for decision-making, the model is augmented further with SHAP(Shapley Additive explanations) to identify the most significant features responsible for accident severity. The model is tested on two real datasets: the US Accident Dataset (March 2023) and the UK STATS19 dataset, following rigorous preprocessing, feature normalization, and class balancing using SMOTE Tomek and RandomOverSampler methods. The suggested CNN–BiLSTM–Attention model presents high predictive accuracy, MAE of 0.137 (US) and 0.179 (UK), far exceeding conventional models such as Logistic Regression. Through SHAP analysis, the framework reveals transparent insights regarding the key factors determining accident severity, and the framework is both precise and interpretable. This paper presents an efficient and scalable solution for real time traffic risk prediction and traffic authority decision support.

Index Terms—Traffic accident prediction, Deep learning, CNN–BiLSTM–Attention, SHAP, Explainable AI, Road safety, SMOTETomek, US Accident Dataset, UK STATS19 dataset, Model interpretability.

I. INTRODUCTION

Road traffic accidents are one of the leading causes of significant human suffering and economic burden in

many parts of the world every year [1].The severity of the injury results from complex factors that include accident factors that affect driver behavior,road conditions,weather and the aspects of the time of day such as rush hour or night driving [2]. Thus,the requirement is to predict the severity of the accident with a reasonable level of accuracy to enhance emergency response strategies and effective proactive traffic management. Conventional approaches like statistical models and traditional machine learning techniques have proved to be inadequate in representing the non-linear and high-dimensional characteristics of traffic data, particularly when spatial and temporal patterns are of utmost importance [3]. Additionally,traditional models tend to be "black boxes" that do not provide the transparency required in high-risk areas such as transport safety [4]. To overcome these limitations, we introduce a hybrid deep learning framework that combines Convolutional Neural Networks (CNN) for spatial feature extraction, Bidirectional Long Short-Term Memory (BiLSTM) networks for temporal sequence modeling, and an attention mechanism to pay attention to the most impactful inputs at decision-making time [5]. The model is also augmented with SHAP (SHapley Additive exPlanations), an explainable AI technique that generates both global and local feature importance insights [6]. This explainable model does not only have good predictive accuracy but also provides clear explanations for its predictions. Identifying the most significant factors causing accident

severity, the model enables data-driven policy formulation and real-time decision-making, ultimately leading to safer and smarter transportation systems [7].

II. RELATED WORK

In recent years, traffic accident severity prediction has emerged as a key focus in intelligent transportation systems. Research has primarily aimed at improving predictive accuracy and model interpretability, though many approaches still exhibit certain limitations. Wang et al. [8] proposed a GRU-AdaBoost model integrating multi-source data (traffic flow, weather, visibility), achieving improved tunnel-specific accuracy, though lacking interpretability. Tian et al. [9] developed a multimodal deep learning model using aerial imagery and segmentation networks (DeepLabV3+, UNet++, SegFormer) to generate intersection risk maps. However, it lacks temporal modeling and explainability techniques. Zepf et al. [10] used traffic simulation to evaluate safety via simulated driving behavior and safety indicators, but did not apply real-world data or predictive modeling. Chandrasekhar et al. [11] applied real-time sensor data and clustering for urban anomaly detection. Yet, their work does not address accident severity prediction or model interpretability. Shim et al. [12] introduced risk grid-based frameworks for AVs using static and dynamic indicators, but these are AV-specific and lack generalizability or interpretability. Thakur et al. [13] presented SafeTrax, an IoT system with GPS and ultrasonic sensors for collision alerts. However, it does not incorporate deep learning or explainability methods. Jeong et al. [14] proposed a gaze-based accident risk prediction system using vehicle-mounted cameras, which is innovative but limited in scalability and structured data usage. Liang et al. [15] used contrastive learning on satellite imagery to predict fatal crash risk. Despite scalability, it lacks temporal and behavioral features for short-term prediction.

III. METHODOLOGY AND EXPERIMENTAL ENVIRONMENT

This section outlines the complete pipeline of the proposed framework, encompassing stages such as dataset preprocessing, handling class imbalance, applying feature normalization, designing the deep learning architecture, and interpreting the model's behavior through SHAP-based explainability.

A. Experimental Setup

The experimental workflow was conducted in a Python-based environment, utilizing both local computing and cloud-based resources to ensure efficiency and reproducibility. Primary model training and evaluation were carried out using Google Colab, which provided access to a CUDA-accelerated GPU environment.

- **Processor:** AMD Ryzen 7 5800HS with integrated Radeon Graphics, 3.20 GHz
- **Memory:** 12 GB RAM
- **Operating System:** Windows 11 (64-bit)
- **Development Environment:** Local development was carried out using Jupyter Notebook and Windows Terminal. Google Colab was employed for training and evaluation using cloud-based GPU acceleration.

B. Datasets

Two large-scale real-world datasets from the United States and the United Kingdom were used to develop and evaluate the proposed model. Both datasets encompass temporal, environmental, and roadway features, supporting robust generalization across different regions. **US Dataset:** Provided via Kaggle (March 2023), this dataset comprises approximately 2.8 million accident records from 49 states collected between 2016 and 2023. It includes weather features (e.g., temperature, humidity, visibility), temporal information (hour, weekday, month), and infrastructure indicators (e.g., presence of traffic signals and junctions). Categorical variables were label-encoded, and accident severity was classified into four levels (0–3). **UK Dataset:** Offered by the UK Department for Transport through the STATS19 dataset, this source spans over a decade of accident records. Accident and vehicle files were merged via a shared identifier. Features include environmental (lighting, weather, surface), behavioral (vehicle type, maneuver), and temporal (hour, weekday, month) data. A balanced sample of 10,000 records was used for training. Severity was encoded into three classes: 0 (Fatal), 1 (Serious), and 2 (Slight).

C. Preprocessing

To prepare the datasets for deep learning, several data cleaning and transformation steps were applied:

- **Datetime Conversion:** Raw date and time fields were transformed into Python datetime objects to enable structured temporal feature extraction.
- **Missing Value Handling:** Features with high missing ratios or low relevance were discarded. Remaining missing values were handled by imputation or row removal based on their impact.
- **Class Balancing:**
 - **US Dataset:** Applied *RandomOverSampler* to replicate samples from minority classes and address imbalance.
 - **UK Dataset:** Used *SMOTETomek*, which combines SMOTE (synthetic oversampling) and Tomek links (under-sampling of noisy borderline samples) to achieve balanced and clean class distribution.

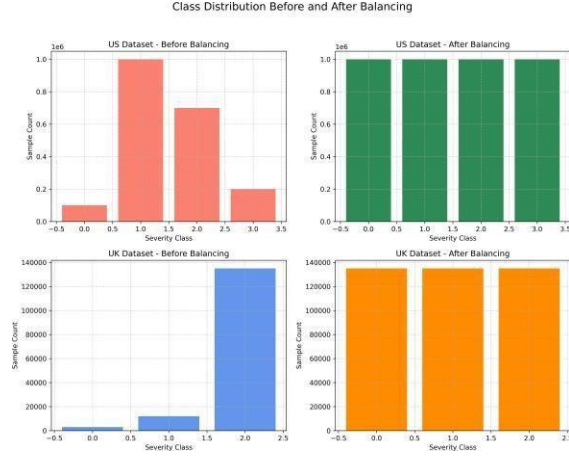


Fig. 1: Class distribution in the US and UK datasets before and after balancing.

- **Normalization:** All numerical features were standardized using *StandardScaler* to ensure zero mean and unit variance. This promotes stable and faster model convergence.

As shown in Figure 1, both datasets had noticeable class imbalance initially. The US dataset had highly skewed severity distribution, while the UK dataset was dominated by one class. After balancing, each class had an equal number of samples, ensuring unbiased learning across severity levels.

D. Feature Engineering

- **Categorical Encoding:** Categorical features such as *Weather Condition*, *Road Type*, *Lighting Conditions*, and *Vehicle Type* were transformed into numerical labels using Label Encoding to allow deep learning models to interpret them.
- **Redundant Feature Removal:** Highly correlated, irrelevant, or duplicate features were dropped to reduce noise and computational overhead.

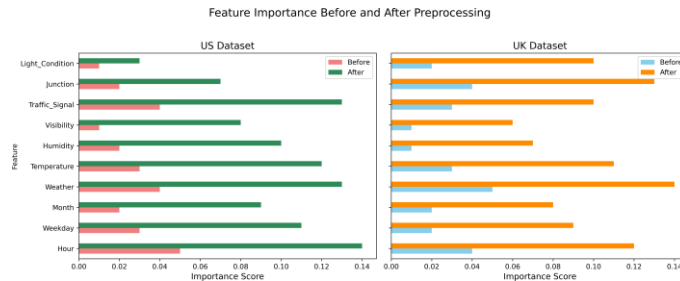


Fig. 2: Feature importance comparison before and after SHAP analysis on US and UK datasets.

These engineering strategies enhanced the quality of input data for the model, improving its learning effi-

ciency and predictive performance. Figure 2 illustrates how feature importance changed after SHAP-based interpretation. Features such as *Hour*, *Weather*, and *Traffic Signal* became more influential, providing insight into key factors affecting accident severity.

E. Model Architecture Components

The proposed CNN–BiLSTM–Attention model integrates spatial pattern recognition, temporal sequence modeling, and attention-based interpretability. [16] Each component plays a vital role in improving prediction accuracy and understanding key influencing features.

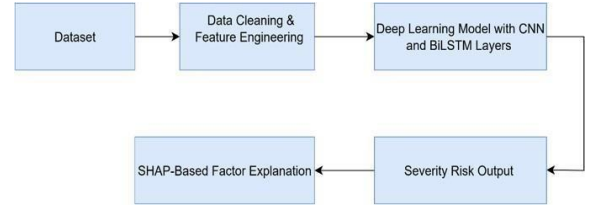


Fig. 3: Workflow of the proposed CNN–BiLSTM–Attention model with SHAP-based explanation.

1) 1D Convolutional Layer (CNN)

The model begins with a one-dimensional convolution layer that treats input features as a sequential signal. This layer captures local spatial dependencies and interactions between environmental and structural variables while reducing dimensionality to retain key patterns. The convolutional transformation is defined as:

$$k-1$$

$$y(t) = \sum_{i=0}^{k-1} x(t+i) \cdot w(i) + b \quad (1)$$

$$\text{ReLU}(y) = \max(0, y) \quad (2)$$

2) Bidirectional LSTM Layer

The CNN feature maps are then fed into a BiLSTM layer that learns temporal relationships from both previous and subsequent contexts in the sequence. This is particularly valuable for determining how accident severity is impacted by the sequence and timing of features such as road type, weather, and time of day.

$$\vec{h}_t = \text{LSTM}(\vec{x}_t, \vec{h}_{t-1}) \quad (3)$$

$$\overleftarrow{h}_t = \text{LSTM}(\overleftarrow{x}_t, \overleftarrow{h}_{t+1}) \quad (4)$$

$$h_t = [\vec{h}_t; \overleftarrow{h}_t] \quad (5)$$

3) Attention Mechanism

To enhance the model's decision-making and interpretability, an attention mechanism is applied to the

BiLSTM output. This allows the model to prioritize significant time steps or feature contributions, highlighting their influence on accident severity. The attention mechanism is mathematically formulated as:

$$e_t = \mathbf{v}^T \tanh(\mathbf{W}_h \mathbf{h}_t + \mathbf{b}_h) \quad (6)$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^T \exp(e_k)} \quad (7)$$

$$\mathbf{c} = \sum_{t=1}^T \alpha_t \mathbf{h}_t \quad (8)$$

4) Fully Connected Output Layer

The context vector from the attention mechanism is passed into a dense (fully connected) layer to produce the final predictions. This layer outputs class probabilities using a softmax activation function:

$$\hat{y} = \text{Softmax}(\mathbf{W}_o \mathbf{c} + \mathbf{b}_o) \quad (9)$$

For the **US dataset**, the model predicts one of four severity categories (0–3), while for the **UK dataset**, the prediction corresponds to one of three classes (0–2), both remapped from original labels.

5) Additional Design Enhancements

Batch Normalization and Dropout are combined to enhance training stability and diminish the possibility of overfitting. ReLU activation adds non-linearity after convolutional layers. The model uses `CrossEntropyLoss` for multi-class classification and the Adam optimizer for efficient convergence. [17] This CNN–BiLSTM–Attention architecture effectively models both long-range and localized patterns in traffic data. The attention mechanism enhances focus on critical features, improving both accuracy and interpretability.

F. Model Training and Evaluation

The hybrid CNN–BiLSTM–Attention(CBA) network was trained over a span of 3 to 15 epochs, adjusted according to the dataset’s size and how quickly the model converged. For multi-class classification tasks, the Cross Entropy Loss function was applied, while the Adam optimizer was chosen for its adaptive learning rate and faster convergence [18]. To maintain a balanced distribution of severity classes in both training and testing phases, we used an 80:20 stratified split. The training process was accelerated using GPU-based computation, and a batch size of 64 was selected to ensure a good trade-off between memory efficiency and computational performance. Model evaluation was carried out using several key performance indicators:

- **Accuracy:** Reflects how many predictions were correct across all classes.

- **Precision (Weighted):** Reflects the proportion of true positive predictions for each class, weighted by class frequency.
- **Recall (Weighted):** Measures how well the model detects actual positives, accounting for class imbalance.
- **F1-Score (Weighted):** Harmonic mean of precision and recall across classes.
- **Mean Absolute Error (MAE):** Computes the average absolute difference between actual and predicted class labels.

Additionally, a confusion matrix was used to visualize misclassifications, particularly between similar classes like *Slight* and *Serious*. Model performance improved with hyperparameter tuning and iterative training. Comparisons with logistic regression highlighted the superiority of the deep model in capturing spatial-temporal patterns.

IV. EXPERIMENT ANALYSIS AND DISCUSSION

This section evaluates the CNN–BiLSTM–Attention model against a baseline Logistic Regression (LR) classifier, focusing on temporal trends, model performance, and interpretability.

A. Comparative Baseline: Logistic Regression

Logistic Regression was trained on the same pre-processed and balanced datasets (SMOTETomek for UK, RandomOverSampler for US). Although it achieved moderate accuracy, it failed to capture spatial or sequential patterns due to its linear nature. In contrast, the proposed deep learning model, enhanced with DeepSHAP, provided both superior accuracy and local interpretability, making it more suitable for traffic risk analysis.

B. SHAP Explainability

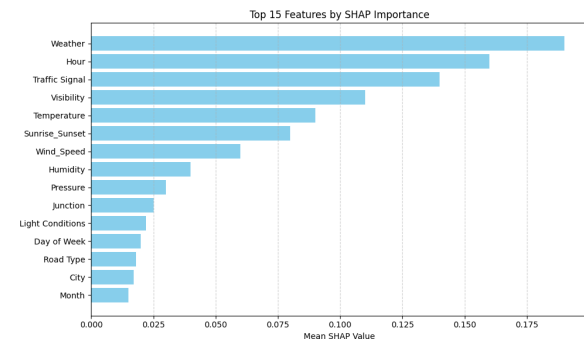


Fig. 4: SHAP summary plot: Top 15 global features

To interpret the CNN–BiLSTM–Attention model, we employed SHAP (SHapley Additive exPlanations) via the KernelExplainer. SHAP assigns importance values to input features, providing both global and local

interpretability. **Global Feature Importance:** Figure 4 shows the top 15 features impacting severity prediction, including *Weather Condition*, *Hour*, *Traffic Signal*, *Visibility*, and *Junction*. **Retraining with Top SHAP Features:** The model was retrained using only the top 15 features, resulting in:

- **40% faster training time**
- **Comparable or slightly improved accuracy and F1-score**

This confirms SHAP’s utility for both model interpretation and efficient training.

C. Temporal Distribution of Accidents

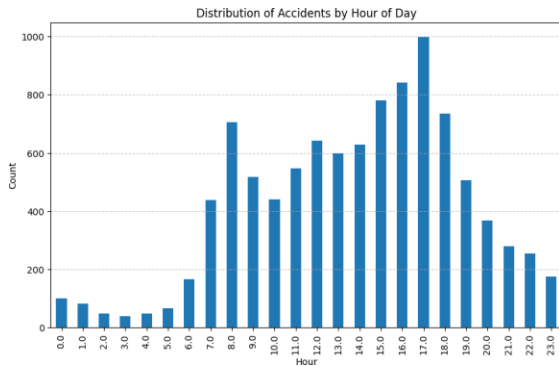


Fig. 5: Temporal distribution of accidents: (a) Day of the week (b) Hour of the day

Figure 5 depicts accident trends across days and hours. Accidents peak during morning (7–9 AM) and evening (3–6 PM) commute hours, especially around 5 PM. Weekdays, particularly Fridays, showed higher accident rates, while Sundays had the fewest—likely due to lower traffic volumes.

D. Performance Evaluation: Confusion Matrix Comparison

As illustrated in Figure 6, the CNN-BiLSTM-Attention model demonstrates strong classification performance with 1464 and 1432 true positives for the ‘Slight’ and ‘Severe’ classes, respectively. Misclassifications are minimal (232 and 255), indicating high precision and recall. In contrast, the Logistic Regression model achieves only 1211 and 1221 correct predictions, with 485 and 466 false classifications. This comparison emphasizes the deep learning model’s superior capability in modeling complex spatial-temporal dependencies present in traffic accident data.

E. Yearly Trends in Accident Severity

Figure 8 illustrates monthly accident severity variation from 2005 to 2016. Severe accidents rose notably

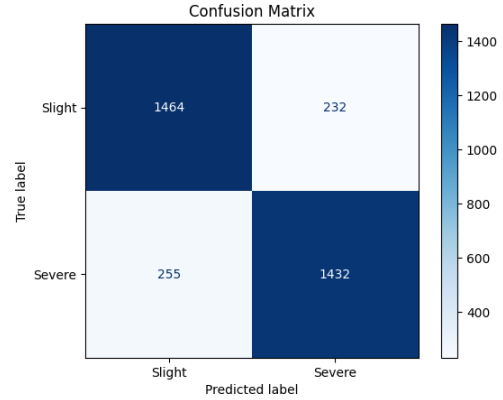


Fig. 6: Confusion Matrix: CNN-BiLSTM-Attention Model

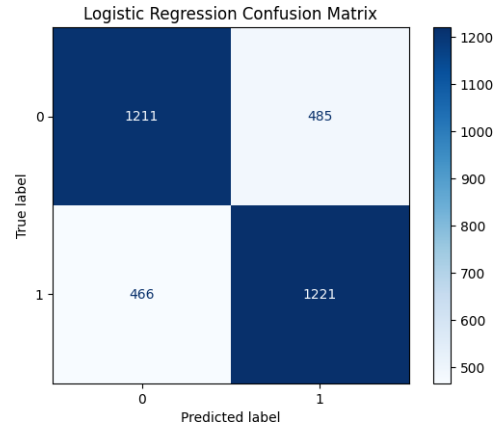


Fig. 7: Confusion Matrix: Logistic Regression Model

between 2008–2009, possibly due to increased traffic or reporting changes. The rise continued after 2013, peaking at over 1000 severe cases in 2016. Meanwhile, slight accidents showed a steady growth, under 200 annually. These findings underline the need for risk prediction and proactive safety policies.

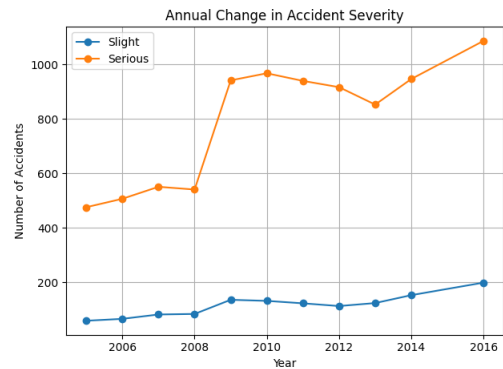


Fig. 8: Yearly Trends in Accident Severity (2005–2016)

F. Spatial Distribution of Traffic Accidents

Figure 9 presents the geographic distribution of accidents across the UK. The map categorizes severity: slight (3), serious (2), and fatal (1). High concentrations appear in southern and central regions, especially around London, Birmingham, and Manchester. Severity 3 (minor) dominates, but scattered severe and fatal accidents indicate the need for localized interventions.

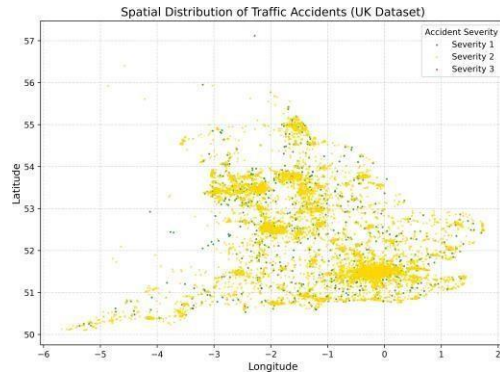


Fig. 9: Spatial Distribution of UK Traffic Accidents

V. CONCLUSION

This paper proposed an interpretable deep learning framework for traffic accident severity prediction by integrating CNN, BiLSTM, and an attention mechanism. The model effectively captures spatial and temporal dependencies in accident data, addressing complex relationships among environmental, temporal, and vehicular features. Experiments on UK and US datasets demonstrated superior performance over traditional models like Logistic Regression, achieving 87% and 86% accuracy respectively, with notable improvements in precision, recall, F1-score, and MAE. The use of DeepSHAP enhanced model interpretability by identifying key factors such as weather, visibility, road type, and time of day. This transparency supports trust and informed decision-making for traffic safety authorities. Overall, the framework combines strong predictive power with explainability, making it a practical tool for real-world accident risk prediction and intelligent transportation planning.

VI. FUTURE WORK

Future enhancements include integrating Transformer based encoders to better capture long-range temporal dependencies and incorporating real-time data streams (e.g., live traffic and weather) for improved adaptability. Deploying the model in live monitoring systems could enable real-time risk prediction and faster emergency response. Additionally, using transfer learning may support

cross-regional generalization. Further exploration of continual learning and multi-modal data fusion (e.g., images, sensors, and text) could expand the model's scalability and effectiveness in broader intelligent transportation applications.

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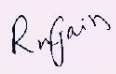
"An Explainable Deep Learning Framework for Predicting Traffic Accident Risk and Identifying Critical Factors Using CNN-BiLSTM-Attention and SHAP"

authored by

S.V.N Sreenivasu, Thirupatamma Kallam, Velangini Rani Biraka, Mythri Yenumala, Neelima G, Malathi Latha T

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